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BACHELOR THESIS

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Arduino platform simulator

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Computer Games Development

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Dedication.

Title: Arduino platform simulator

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Abstract: This thesis focuses on simulating code of Arduino microcomputers in a virtual environment with a graphical user interface without the need of emulation. An important part of the thesis shows how and why should be the simulator and the GUI split into two processes, and how to keep them synchronised. This introduces the reader into the field of interprocess communication and its implementations using TCP connection, shared memory and events. Another important part focuses on simulation of digital circuits with extensible component library. It describes how new components and circuits can be created from YAML, SVG and C# files by the user.

Keywords: digital circuit simulator, Arduino

Název práce: Simulátor prostředí Arduino

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Abstrakt: Tato práce se zaměřuje na simulaci kódu mikropočítačů Arduino ve virtuálním prostředí s grafickým uživatelským rozhraním bez nutnosti emulace. Důležitá část práce ukazuje, jak a proč by měly být simulátor a GUI rozděleny do dvou procesů a jak je udržovat synchronizované. To čtenáře uvádí do oblasti meziprocesové komunikace a její implementace pomocí TCP spojení, sdílené paměti a událostí. Další významná část se zaměřuje na simulaci digitálních obvodů s rozšiřitelnou knihovnou komponent. Popisuje, jak může uživatel vytvářet nové komponenty a obvody ze souborů YAML, SVG a C#.

Klíčová slova: simulátor číslicových obvodů, Arduino

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Introduction

Arduino is a very popular microcomputer among amateurs and professionals alike. Despite this, testing the code for an Arduino machine can be fairly difficult. Debugging options are limited and the user needs to rely on additional measuring equipment. The usual choice is to test the code with an AVR emulator. Emulation is the process of running machine code on a hardware that doesn't match its instruction set. AVR is the processor that controls the Arduino board. These emulators are sometimes part of a bigger logical circuit simulation system, other times they are just a single piece of software.

Goal of this work is to skip the emulation process and create a working Arduino simulator by mimicking the changes of digital values on the board's pinout. The user code will be compiled directly to the x86-64 architecture instruction set, which is our target platform, and thus able to run at the native speed of the host machine.

Bundled with this simulator is a graphic application for creating and testing logical circuits. This GUI is written with Avalonia framework and will contain Arduino as it's main component. The point of this application is to attach to the running Arduino simulator and visualize the expected Arduino behavior with user's code on a given circuit. We want to give the user the power to compose their own circuits, and define the behavior and looks of their components. Additionally, the user will be able to debug their program, while the GUI is running, using their preferred IDE and C++ debugger, providing real-time feedback and inspection when stepping through the code.

Structure of the thesis

This thesis starts with a brief introduction of technologies we've chosen for achieving our vision from the introduction. Then we lay down very specific goals and requirements, we want to address in this very thesis, using the mentioned technologies. In chapter 1 we introduce the reader to programming for Arduino boards. We show what is required from Arduino programs (1.1), how they access the board's I/O (1.2), and how are they built (1.3).

In chapter 2 we provide a high level look on the core systems of the work. We show why it is required to split the work into 2 applications (or more accurately processes) and we expose the reader to the world of inter-process communication (2.2). We show how it can be achieved using TCP connection (2.2.4), shared memory (2.2.5) and events (2.2.3), and what kind of problems had to be taken into consideration (2.2.1). Then we delve into more details on how the individual projects and classes have been implemented in chapters 3 and 4.

Finally, the chapter 5 consists of specifics on how to simulate Arduino code using the applications we developed for this thesis (5.1). We specify how to create custom components and scenes for the circuit simulator (5.2). And in the very end, we specify what interprocess communication protocols we have created for interacting with the simulator's backend, so that any other user interface can potentially be connected to it (5.3).

0.1 Technologies used

C++, CMake, Boost

For technical reasons described in the subsection 2.2.1, the work is split into two processes: the frontend and the backend. The backend is written in C++, as it is almost identical to the language used for Arduino sketches (see 1.3). This makes the build process of the user's code much simpler. In the C++ project we use Boost libraries that provide useful abstractions for communicating with other processes (`asio`, `interprocess`) and parsing command line arguments (`program_options`). We use CMake as our metabuild system of choice, as it is the one we are the most familiar with.

C#

The frontend is written in C# from the .NET 9 Software development kit. We consider the language to be more convenient for writing many of the essential systems than its C++ counterpart. That includes handling the graphics, the user interactions, abstraction and parsing numerous configuration files. We also believe the language will be more comfortable for users that wish to extend the simulator with their own components. Libraries are also much easier to set up and compile thanks to NuGet.

Avalonia UI

For the GUI we decided to use Avalonia UI. Avalonia is a popular open-source cross-platform .NET framework, used by major companies like JetBrains or Unity. It is based on the Microsoft's WPF framework utilising the same XAML markup language and many of the same core principles. We chose it specifically because neither WPF or its cross-platform successor MAUI are available on Linux, one of our target platforms.

SVG

We want to render the circuit's components using vector graphics, as they don't deform, no matter how much scaled they are. This is crucial, since we also want to render the components in different sizes and under different zoom levels. The most common file format for vector graphics is SVG, so this is format we will require supplied for custom components. Avalonia provides a sufficient support for rendering vector graphics using SkiaSharp under the hood, as all of its built-in controls are rendered using this framework.

0.2 Specific goals of the thesis

Target platforms:

- Linux (x86-64)
- Windows 10 or higher (64-bit)

Software prerequisites:

- `arduino-cli` ¹
- Minimal system requirements of Avalonia UI ²
- .NET 9 SDK or higher
- C++ compiler
- CMake
- C++ Boost libraries:
 - `asio`
 - `interprocess`
 - `program_options`

Simulator functional requirements

- Defines platform independent constants and macros from `Arduino.h`.
- Implements `digitalWrite`, `digitalRead`, `pinMode` and `millis` functions from `Arduino.h`.
- Declares and calls `setup` and `loop` functions with the behavior expected on an Arduino machine. ³
- Compiles and runs any valid Arduino UNO R3 SMD sketch, preprocessed with the `arduino-cli` tool, containing only symbols mentioned above.
- Reflects behavior of the above functions in an internal Arduino state. Logs changes of this state in a text file.
- Integrates an interprocess communication mechanism for exchanging data with an external GUI application.
- The events incoming from a GUI application must be resolved in a manner, that preserves the continuity of events.

GUI functional requirements

- The application must stay responsive during debugging of the simulator process.
- The application must be able to connect to the interprocess mechanism of the simulator.
- Simulates simplified digital circuits. Exact physical properties, like voltages and currents might not be implemented.

¹See section 1.3

²See <https://docs.avaloniaui.net/docs/supported-platforms>

³The significance of the mentioned functions is described in detail in the chapter 1

- Implements mechanism for defining custom components and visuals with `.yaml`, `.cs` and `.svg` files. These components include interactive controls, such as buttons.
- Implements mechanism for defining a circuit (scene) using the custom components.
- Predefines LED, button, voltage source and Arduino components, and a scene which utilizes these components.
- The changes to the simulator's internal state must be immediately reflected on visual components connected to the Arduino in the scene.
- The scene is able to be zoomed and panned. The visuals of the components must be rendered using vector graphics.
- Multiple interactable components may be pressed at once.

1 Introduction to Arduino programming

1.1 Sketches

The Arduino programs are commonly referred to as *sketches*. A sketch is not a single file, but an entire directory, usually populated by files with the `.ino` extension, containing the source code of the program. `.cpp`, `.c`, `.S` (assembly files), and header files like `.h` are also accepted as files containing source code. Every sketch must contain a `.ino` file with a file name matching the sketch root directory name [1].

The Arduino programming language, in which `.ino` files are written, is a C++ dialect with very subtle differences, mostly concerning linking of files and omitting function declarations. Bodies of function definitions and variable declarations can be considered as a perfectly valid C/C++ code.

The essential functions that an Arduino programmer has to define in their sketch are `void setup()` and `void loop()`. The `setup()` function is called once, when the program is booted. That is when the Arduino board is powered up, the program is uploaded to it, or when the reset button is pressed. The `loop()` function is called in a loop indefinitely. This is the proper place to define any interactions with the board, detecting inputs, updating timers, etc.

1.2 Arduino board interface

The target board `Arduino Uno R3 SMD` provides 6 analog and 14 digital general purpose pins, some of which can be used for I2C and SPI protocol communication, or PWM modulation. The most simple to use and simulate are the digital pins for general purpose I/O, so this thesis focuses exclusively on using these pins as such. [2]

The digital pins can be manipulated using only these 3 functions:

`void pinMode(pin, mode)` configures specified pin either as an input or an output

`value digitalRead(pin)` reads the voltage on the pin, stores it as a digital value, either 1 or 0

`void digitalWrite(pin, value)` when set to output, sets the voltage of the pin to corresponding value.

Pins are identified by their integer index. `mode` and `value`'s underlying type is also an integer, but it is preferred to use predefined constants, such as `INPUT/OUTPUT`, `HIGH/LOW` and such. Below is an example of a sketch from the Arduino's documentation, using only the described functions:

Listing 1.1 Sketch example [3]

```
1 | int ledPin = 13;  // LED connected to digital pin 13
```

```

2 int inPin = 7;      // pushbutton connected to digital pin 7
3 int val = LOW;     // variable to store the read value
4
5 void setup() {
6     pinMode(ledPin, OUTPUT); // sets the digital pin 13 as output
7     pinMode(inPin, INPUT);   // sets the digital pin 7 as input
8 }
9
10 void loop() {
11     val = digitalRead(inPin); // read the input pin
12     digitalWrite(ledPin, val); // sets LED to the button's value
13 }

```

1.3 Build process

Before the code is compiled, all `.ino` go through a simple preprocessing stage. `#line` directives are added. These link to the original line and file, when the code is debugged. All files are concatenated together, function declarations are generated at the top, and the `Arduino.h` is included. This header file is crucial as it declares the `void` and `loop` functions, functions for manipulating pins, common constants and mathematical functions, etc. At this point the sketch becomes a complete C++ source code.

It is also at this where we steer away from the standard compilation process, and apply our own `Arduino.h` implementation in our simulator. Usually, the C++ code would be linked to an implementation that contains binaries and constants for the given microchip installed on the target board. Then it would be compiled with a compiler for the given instruction set (e.g. `avr-gcc`). The code would be then transmitted over USB to the user's Arduino [4].

In our case, we include the entire preprocessed sketch in our simulator project written in C++, and compile it as whole for the instruction set of the host machine. This executable will implement mock functions declared in the `Arduino.h`, arduino's internal state, and a way to communicate the change of state to an external GUI application.

The entire process from preprocessing to compilation, can be orchestrated with the use of `arduino-cli` command line application. We will use it to preprocess the `.ino` for us, so that we end up with a clean C++ file we can include in our project.

2 Architecture design

2.1 Architecture overview

The whole project is divided into two main parts. The purpose of the backend is to execute the Arduino code and simulate the change on the board's pinout. The purpose of the frontend is to reflect these changes on a user defined logical circuit, while offering the user control over the simulation. We chose to implement the backend in C++ and the frontend in C#. Both backend and frontend are running in its own separate process.

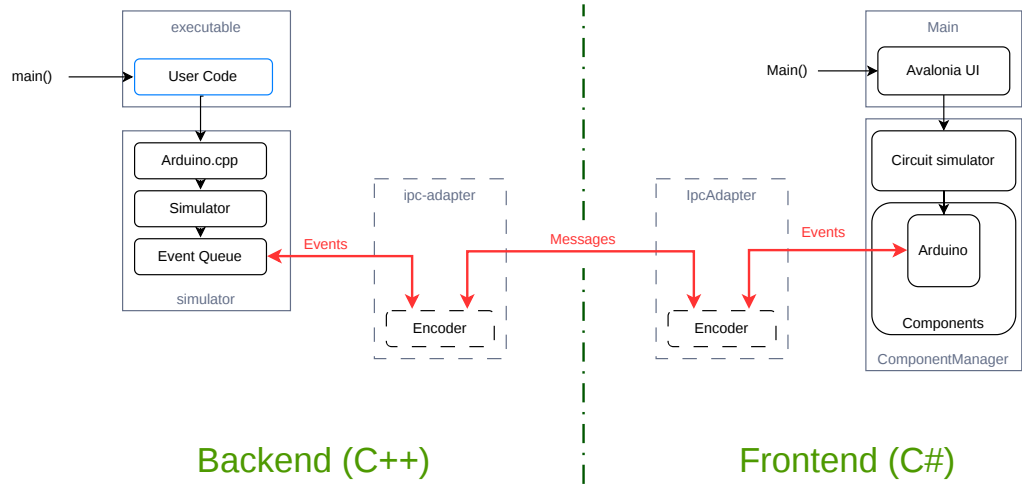


Figure 2.1 Top-level architecture overview

The Figure 2.1 shows the most important systems in each process. The user code is compiled together with the **executable** module of the backend. The declarations in the header **Arduino.h** prepended to the top of the **.ino** is implemented by a mock library **Arduino.cpp**. For each observable action an event is created and passed to the singleton **Simulator** class to compute and put into an event queue. The event is also communicated to the frontend process.

In the frontend the events are consumed by the **Arduino** component, one of several components predefined as part of the **ComponentManager** project. This project is responsible for loading user-defined components and scenes, which connect the components together in a circuit. The circuit is simulated in the same project. The user interface is implemented in the **Main** project using Avalonia UI as a framework. Interacting with the UI may invoke new events in the **Arduino** component. Those are sent back to the backend's event queue in the same manner as they were received.

The IPC Adapter is an interface implemented in each process responsible for encoding and decoding the events into a standardised format and relaying to the other process.

2.2 Inter-process communication

Implementing processes that depend on each other comes with certain challenges. Processes always run concurrently, so a variety of synchronization techniques must be applied to avoid race conditions. Before that we need to determine how will the processes exchange data. Process is a kernel level mechanism, thus approaches to handle the exchange can vary between operating systems. These mechanisms are what is called *Inter-process communication*, or IPC for short.

2.2.1 Considerations

There is a technical and an engineering reason for splitting the work into more than one process.

The user must be able to have full control over the frontend user interface at all times, even while the Arduino code is being debugged. Every common debugger blocks all running threads of a process, when a breakpoint is hit. **elaborate** If the GUI has been part of the same process as the systems running the simulation, it would freeze and become unresponsive every time a breakpoint is hit. The only way to avoid this behavior is to separate the GUI to a different process. We want to support breakpoints in our project, as they are the most useful tool in a debugger's toolkit.

Splitting the application into multiple processes is actually a common practice among larger pieces of software. The program of each process can be written in a different programming language and thus delegated into teams with different competencies. They mark clear separation of concerns and force a solid interface design. Each process is easily swappable for another implementation with the same interface while the other processes are still running. In our case, the GUI process could be replaced by a web client, a CLI client, or an IDE add-on.

There are more than a few ways to handle IPC on every OS. The ones that we implemented are *TCP stream* and *shared memory*. These are among the most common approaches available both on Windows and Linux. In either case, we must clearly define what the shared data represent. Although shared memory has more flexibility than TCP, we stuck to sending discrete text or binary messages of predetermined format over time, to keep the implementation simpler and the two approaches interchangeable.

Another question is what will we encode in our message. At first we encoded the full state of the simulated Arduino in *snapshots* but due to its various shortcomings we later switched to sending discrete *events*. So the simulator in the backend and its interpretation in the frontend both operate in terms of events, but TCP and shared memory modules only understand the notion of sending and receiving messages with very distinct interfaces. That's why we need to design an adapter in both processes and for both IPC methods that sets up the mechanism and encodes or decodes the passing messages into structured event data the simulation logic understands. The IPC adapter then informs the other systems in it's process any time a new event is received.

2.2.2 Snapshotting

In the early iterations, a snapshot of the Arduino state was sent over one message. That included the pinout state, pin modes, timer, etc. The state was periodically requested by the client with a *Read* message without payload. When a new value is read on an input pin in the frontend, a *Write* message with the state as payload would be sent to the server and the values of input pins would be overwritten atomically to the backend's internal state.

This approach was easy to implement but flawed. The main issue was the frequency of reads. At lower frequencies, the system is prone to temporal aliasing. **Find temporal aliasing definition.** Some events, like very short impulses, can be missed completely. This is not much of an issue in purely combinational circuits. But we want our frontend to support stateful components, e.g. shift registers present on the Funshield, which are controlled by digital writes less than a microsecond apart.

Highering the frequency to such an extent is not viable. **elaborate**

2.2.3 Events

Right now synchronization between processes is maintained by recording every significant event produced by either the backend or the frontend, and sending it to the other process. Each process is equipped with an event queue, that consumes the events in order of the least recent to the most recent. Consuming an event means invoking the appropriate function based on the type of the event with arguments stored in the event's signature.

The Arduino object in the frontend has no logic on its own, so the only consumed events are those produced by the backend. For that reason a simple queue is sufficient, and no further synchronization is necessary. In the backend however, both simulator's and frontend's events are consumed. Sending an event to the frontend and receiving direct consequences of the event back can involve a significant time overhead. In the meantime the simulator produces more events that are consumed almost instantaneously. We need to ensure that the events are consumed in a valid order. We also need a robust queue implementation that isn't prone to race conditions.

An event is labeled with two timestamps. Timestamps don't have to be based on a real clock, the only requirement is that it is a non-decreasing sequence of numbers. The first timestamp sent is the simulation time, the time as perceived by the Arduino being simulated. That is time given by a clock that is started right before the `setup()` is invoked. It can be manipulated by the user by pausing or slowing down. This is to aid the user while analyzing the logs of the simulator.

The other timestamp included is *Local virtual time* (LVT)¹. Each process holds its own integer value of LVT. This number is increased with newly produced events and events received from the other process. When a decision has to be made on which event to consume next (like in the case of the backend's queue), the event with lower LVT is chosen. The relation $LVT_1 > LVT_2$ means that the event with LVT_1 happens after the event with LVT_2 . In other words, event 1 is the consequence of the event 2 regardless of the real time passed between the events.

The rest of the information stored in an event is its type and arguments. The type corresponds to the action taken after the event is consumed, like writing to a pin or changing pin mode. Each type can define up to 3 integer parameters, e.g for a *write* event that is id of the pin, and a digital value we want to write to it. By limiting the number of arguments, we intend to avoid heap allocation of these events, which could considerably slow down the simulation with programs that produce a lot of events in a short time.

There are many ways to encode such an event into a message. Currently the only encoder implemented in this work is the text encoder, which parses values of events into human readable strings separated by a colon. The messages themselves are separated by a semicolon. This particular encoding is quite space ineffective, but it helps a lot during debugging of the IPC systems.

2.2.4 TCP stream

Messages can be sent and received through a stream of data on a TCP connection. TCP is commonly used to send data over a network, meaning backend and frontend processes can run on different machines. But listening to a connection on the same machine on the same port works just as well. TCP messages are guaranteed to arrive in the same order as they were sent.

The backend implements the TCP server, while the frontend implements the client. The implementations and design considerations behind these are very different based on the language they were written. But in both cases TCP handlers need to run on their own threads, due to how heavyweight **elaborate** they are. There is a buffer to which read bytes are stored from an OS network stream. These are passed as a chunk to a higher level message handler that splits the chunk into individual messages according to predetermined delimiter character. When a message needs to be sent from another thread, it is passed to the message handler where it is posted to the end of a queue and is scheduled to be processed by the OS.

There is a minor limitation of IPC over TCP that makes the debugging experience of the user's code uncomfortable. As we established, hitting a breakpoint will block all running threads of the process, including the thread with the TCP handler. When stepping through the lines of the code where events are produced, the TCP thread has to schedule posting of the message with the event, and it takes time for it to be eventually sent. In many cases the TCP thread won't be fast enough to send the message in the time the main thread progresses to the next line where a breakpoint will be hit again. As a result, consequences of the event won't be reflected in the frontend process until another one or more lines have been stepped through. This is the main reason we decided to also implement IPC with memory mapping, even though TCP is fully functional.

¹The terminology and principles are inspired by the *Optimistic synchronization* paradigm in distributed systems. In this paradigm, the events are consumed as fast as possible, and then rolled back once an event with a lower LVT than the ones consumed appears (so called *Time Warp* mechanism). But Time Warp is not viable for simulating code of a high level language like C++ without custom memory management, and in our implementation it isn't even necessary. [5]

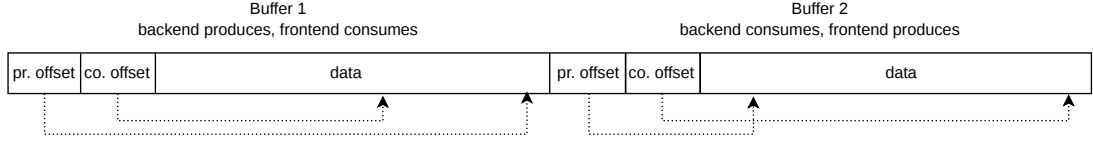


Figure 2.2 Memory layout of the shared memory region

2.2.5 Shared memory

Memory mapping is the functionality of an operating system to relate a region of main memory 1:1 to a region in the address space of a process. This is an important idea in establishing a virtual memory of a process, but OS objects can be mapped as well. It could be either a *memory-mapped file*, contents of a disk file moved to the RAM. Or it could be a special OS object called *shared memory*, living in the kernel heap, specifically designed for IPC. In both cases, multiple processes can map the same memory region to their own address space. Thus they have access to the same data.

The terminology and API differ significantly between Linux and Windows, but with the use of the right libraries, the differences can be abstracted away and used as a single concept. So from now on we refer to the idea of mapping the same memory region to multiple address spaces as *shared memory*.

This tool is very powerful, we can address the shared data by pointers as we would with any other data of the process. There is also no need for system calls once the the memory has been mapped. No special instructions **atomic??** need to be used for reading or writing and all changes are immediately visible on all of the processes that map the same region. As long as we do the writes in the simulating thread, we avoid the problem with delayed observations during debugging we encountered with TCP.

In the shared memory region, we establish a data structure that stores messages in a sequential order. We opted for a circular buffer to fully utilize the limited space of the region. There are actually two non-overlapping buffers in the same region together with a few bytes of metadata about their current state. A process consumes messages from one buffer while producing new messages to another. This is to avoid numerous race conditions that would occur while writing and reading from the same buffer while keep the synchronization mechanism lock-free.

As for the metadata, for each buffer a position after the last consumed and produced message is stored. It is important to never store references as pointers in the shared memory itself. The memory region will be most likely mapped to different addresses in each process. Because of that the *producer offset* and *consumer offset* are regular integers denoting the distance from the first adress of the buffer in bytes. When the offsets are equal, it means the buffer is empty. We don't care about concurrent updates of these offsets, one process changes only one of the offsets for the duration of the simulation.

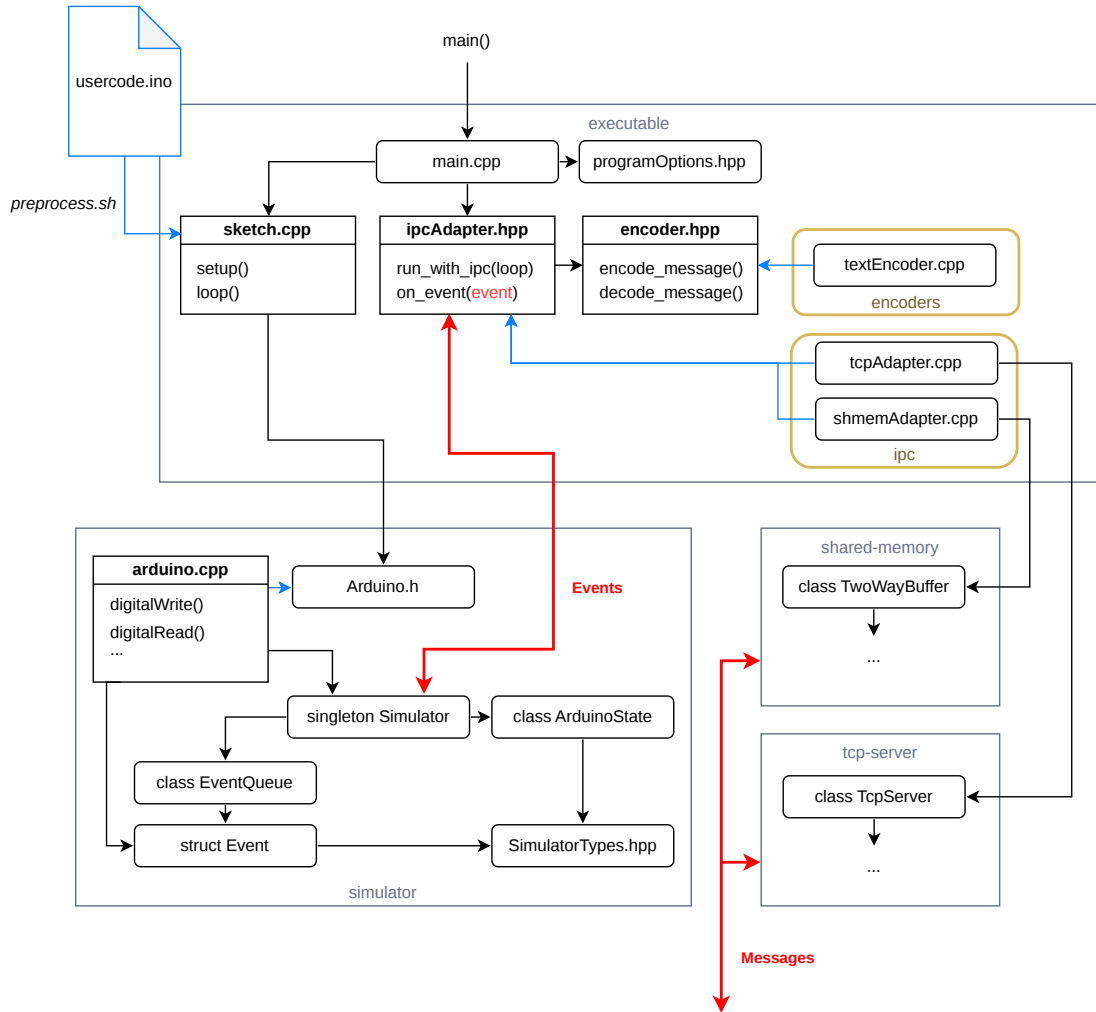


Figure 2.3 Backend architecture

2.3 Backend

2.3.1 The simulation

The language choice for backend was obvious, the `.ino` file is a syntactically valid C++ code. We include the sketch source code in the project by passing it as an argument to the `preprocess.sh` script (`preprocess.bat` on Windows). Inside we use the `arduino-cli` tool with the right arguments to apply very light preprocessing to the file, like adding function declarations to the top or including the `Arduino.h` header. Important additions are the `#line` directives, which link to the definitions in the original `.ino` file. Thanks to these, breakpoints can be added to the `.ino` file and debugged as if the file itself was part of the project, completely with stepping through lines and inspection of variables. The file is then saved as `sketch.cpp` in the directory of the `executable` module where it is picked up by the CMake configuration and included in the `main.cpp`. The functions declared by the `Arduino.h` header file, like `digitalWrite` or `millis` are implemented in a custom mock library `Arduino.cpp` in the `simulator` module. Instead of instructing the AVR chip, as the real implementation would, the mock definitions create events and pass them to the `Simulator` for enqueueing and

consumption.

The `Simulator` singleton class is the beating heart of the backend. It is responsible for handling the events and the event queue, managing the Arduino state, logging changes and keeping track of time. It is initialized in `main.cpp` and used frequently throughout the `executable` and `simulator` modules. The `Simulator` holds the instance of the `ArduinoState` class, representing the digital states on its pins, current pin modes and the inner timer of the Arduino. It also holds the instance of the `EventQueue`, which stores both local and remote events waiting for consumption. The `SimulatorTypes.hpp` is a header file with very few type definitions commonly used in the `simulator` module.

2.3.2 Events and IPC

The `Event` struct is just a "recipe" for what changes should the `Simulator` do to the `ArduinoState`. It holds a lambda function, corresponding to an event's type that is executed only after the event is put in the event queue and consumed in the order described in the section 2.2.3. The struct defines static functions that create new event instances with these predefined lambdas with the event type's parameters. When a local event is consumed, i.e. the lambda is executed, the event is signalled back to the `executable` module through a callback, that has been passed to the `Simulator` during initialization. This way the event can be sent to an IPC adapter, regardless of the used implementation, without a circular dependency on the `executable` module.

The IPC adapter itself is an interface of two functions declared in the `ipcAdapter.hpp`. `run_simulator_with_ipc` initializes the IPC mechanism. It takes the simulator loop function as an argument and executes it after IPC has been set up. That is because the IPC usually needs to create new threads that cannot be joined before the loop stops running. The loop function itself is defined in `main.cpp` and it contains both the `setup()` and `loop()` calls from the user's code.

The `on_event` function of the adapter is the callback passed to the `Simulator`. It encodes the event into a message and sends over IPC to the remote process. Listening to events of the remote process occurs on a different thread than the one with the simulator loop. Incoming messages are decoded into `Event` structs and sent directly to the event queue in the `Simulator`. `encode_message` and `decode_message` are functions declared in yet another interface in the `encoder.hpp` header. It's implementations are in the `encoders` directory, i.e. only `textEncoder.cpp` at the moment.

Implementations of the IPC adapter are located in the `ipc` directory. The `tcpAdapter.cpp` depends on the `tcp-server` module, while the `shmemAdatper.cpp` depends on the `shared-memory` module. Which one is used depends on the chosen compilation target. The `tcp-server` module contains definition of the TCP server and mechanisms for accepting new connections and handling the stream of data using the `boost::asio` library. The `shared-memory` module contains memory mapping mechanisms using the `boost::interprocess`, implementation of the circular buffer and the producer-consumer pattern using two buffers.

2.3.3 Utilities

Logger

There are additional utility modules not shown in the figure 2.3. One of them is the humble **logger** used extensively throughout the project. The **ILogger** abstract class exposes the **log** function accepting either a raw string or an **IMessage** abstract struct. Structs inheriting from **IMessage** represent unified formatting of a message with a timestamp and a level of verbosity. The verbosity levels from the highest to lowest are *Debug*, *Info*, *Warning* and *Error*. The messages of higher verbosity levels can be omitted from the logs by setting the appropriate command line option. **ILogger** offers shorthand functions for logging these general messages, but more specific ones can be defined as well.

The module comes with **FileLogger** implementation of the **ILogger**. It creates or opens a file and dumps the logs inside. To avoid race conditions while writing to the files, two of these loggers are used in the project. One is owned and used by the **Simulator** singleton for logging inner workings of the Arduino. The other is owned by the IPC adapter and used for logging the exchanged messages or potential failures of the IPC mechanisms.

Timer

The other utility model is **Timer**. It is used to track time from the start of the simulation. The simulator holds two of these timers, one for *real time* and one for *simulation time*. Although there is no difference between them right now, the intention is to apply effects like stopping or slowing down to the simulation time timer from the frontend, so that functions like **millis** work consistently with disturbances caused by the frontend. Functions returning real time and simulation time are passed to the file logger as a source of its timestamps.

2.4 Frontend

2.4.1 The presentation

Our humble GUI is written using the Avalonia UI framework. It consists mostly of the circuit canvas, a zoomable and pannable space where the *components* of a circuit can be placed to specific coordinates, unconstrained by the usual layout structures of Avalonia. Arrangement of these components is called a *scene*. Each of the component types have its own set of SVG images that are drawn to the canvas. We refer to them as *sprites*. Each individual component always references one of its sprites, conveying the current state to the user. A sprite on the canvas can be clicked on (e.g. a button) to generate some kind of response. Of course, these features are not built into Avalonia, so we need to create custom Avalonia *controls* to achieve them. We only include Avalonia in the Main project of the frontend, as this is the only project responsible for rendering.

Model-View-ViewModel

The preferred architecture for Avalonia applications is *MVVM*, or *Model-View-ViewModel*. *View* is the visual layout of the window and its parts. It is written

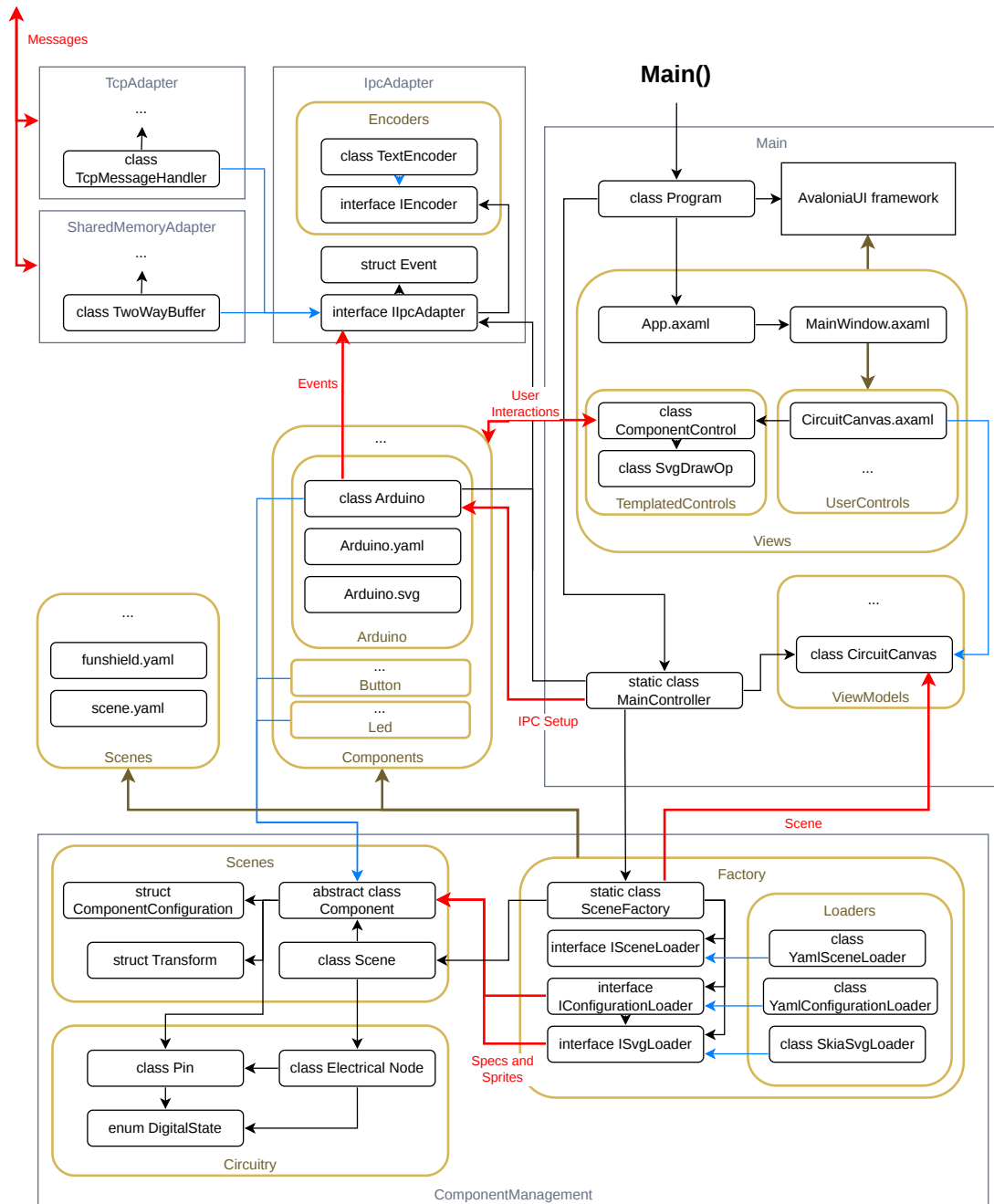


Figure 2.4 Frontend architecture

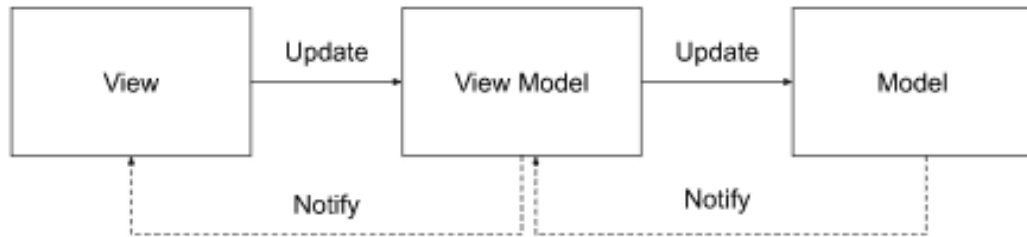


Figure 2.5 MVVM architecture [7]

in *XAML* (Extensible Application Markup Language) in files with the `.axaml` extension. In Avalonia’s case, elements of a view are called *controls*. Those can be buttons, labels, containers, etc. 3 types of custom controls can be created in total, our frontend utilizes *user controls* and *custom-drawn controls*. User controls are themselves views built up in XAML from other controls. This is perfect for our `CircuitCanvas` view, as it is just a list of component controls enveloped in a built-in `Canvas` panel. The custom-drawn controls are defined as C# classes, inheriting from the `Control` class, and defining it’s own rendering logic by overriding the `Render` method [6].

We need to implement the visuals of our components using custom-drawn controls, as there are no built-in or library controls that can draw a precompiled SVG image and swap it during runtime. Thus we define a `ComponentControl`, that calls a custom drawing operation in its `Render` method. That is defined by the `SvgDrawOp` class, which takes a sprite and a transformation matrix as arguments, applies visual transformations on the image and instructs the Avalonia’s rendering pipeline to show it in the parent view.

XAML views bind to data and events in a *viewmodel*, which manipulates with data shown in the view. A model then is the raw data passed to the viewmodel, accessed by some database or a service, or produced by the business logic. View is aware of the viewmodel and viewmodel is aware of the model, but not vice versa. In ideal case, the model should be agnostic about the UI technology used. [7]

Our model is the abstract `Component` class implemented by various types of components, like LEDs, buttons or our Arduino. A list of the `Component` instances is stored in `CircuitCanvas` viewmodel, and each instance is passed to a `ComponentControl` in the `CircuitCanvas` view. This and all other initialization steps of the application are orchestrated by the static `MainController` class.

2.4.2 The logic

Only a minor part of the frontend is actually written with Avalonia. The business logic of the aforementioned `Component` model is all encompassed in the `ComponentManagement` project. That means parsing of the components definition, creating instances of a scene, and the simulation of the circuit. This project is completely unaware, that it will be rendered with Avalonia specifically.

The circuit can be thought of as an unoriented bipartite multigraph, where nodes of one partition are *electrical nodes* and nodes of the other partition are *components*. An edge of the graph represents a conductive connection between an

electrical node and a *pin* of a component. Components usually have multiple pins.

In our digital circuit, an electrical node always obtains one of two digital states, *high* or *low*, which is given by states of the pins it is connected to. When a high state is written into a pin, we say it is *driving*. When a low state is written into a pin, it becomes not driving again. The digital state of an electrical node is high precisely when at least one of the pins it is connected to is driving.² A change in the state of an electrical node is propagated through all of the connected pins. Components can react to this change by writing new states to their other pins creating a chain reaction of state changes in the circuit.

Components

The frontend's user is encouraged to extend the simulation with their own implementations of the abstract **Component** class. That is done by creating a new *component directory* with the name of the component. The shared path to all component directories is specified as one of the command line options of the frontend. The static information about the component type is stored as a record **ComponentConfiguration**, created from the files in the directory. The required files are a YAML file with metadata, a file with C# class, and at least one SVG file for the sprite. The YAML file contains in the very least a name to identify the component type, and a list of the component's pins. The pins can be defined either as an enumeration of their names, or as a single integer, that represents the total count of unnamed pins identified by their index. For full specification and an example of the configuration's YAML file, see subsection 5.2.2

The **Component** base class provides multiple virtual methods with the **On*** prefix. These are called upon changes in the circuit or as a result of a user interaction. The most important one is the **OnPinStateChanged** with the **Pin** object passed as an argument. This is the primary way of defining the component's logic. **OnControlPress** and **OnControlRelease** are other useful methods for interactible components, such as buttons. The base method **UpdateSprite** changes the rendered sprite to a different one, identified by the name of its SVG file. The base **GetPin** method returns a **Pin** object of the component's instance identified either by its name or index. Digital states are read and written through this object.

Usually, specific pins have inherent semantics of being used only for reading or only for writing. This property can be set explicitly on the **Pin** object with **MakeReadOnly** and **MakeWriteOnly** methods. Read only pins refuse write attempts from the component's logic and write only pins refuse notifying a component with a new state from node's propagation. General pins, defined by method **MakeGeneral** don't have these constraints. Incorrect use of pins with these semantics is followed by a warning in the simulation logs.

²This is a simplification of how the term *driving* is usually used. *Drivers* are generally devices that force a state or signal on the electrical node. That could be both high or low digital states, or it could be e.g. a battery driving an LED. Driving both low and high state to a node may lead to short circuiting, which is a situation impossible in our simulator. For more specific example, see how VHDL standard defines driving outputs. (Annex A.1, A.9 [8])

Scenes

The final form of the graph is achieved with creation of the scene. The YAML file of the scene consists of two sections. In the **Components** section, individual instances of **Component** subtypes are defined. They must have a unique name, and they should specify a graphic transformation, i.e. position, rotation and scale. These values are composed into a **Transform** struct and passed into the **SvgDrawOp** as a transformation matrix. The **Nodes** section defines individual electrical nodes. A single node in this file is represented as a list of component-pin pairs, where the components are identified by their instance name in the **Components** section, and the pin is identified by its definition in the **ComponentConfiguration** of the given **Component** subtype. For further details see the subsection 5.2.4.

The creation of the scene is comprised of a series of parsers in the **Factory** directory. There is an interface for each of the file types, i.e. the scene, component configurations and SVG files. Each has exactly one implementation in the **Loaders** directory, using **YamlDotNet** and **SvgSkia** libraries. They are put together in the static **SceneFactory** class, where component subtypes and instances are finalized. The results is a complete **Scene** object with initialized **Component** instances and **ElectricalNode** lists. The **SceneFactory** is called by the **MainController** in the **Main** project, which passes the scene to the **CircuitCanvas** viewmodel.

2.4.3 The Arduino component and IPC

Arduino is one of the component subtypes of the **ComponentManagement** project. It is a partial class divided into 2 files. In **Arduino.cs** it overrides virtual methods and deals with interaction with the circuit. In **Arduino.Events.cs** it creates events and sends to IPC. It also calls its own method from **Arduino.cs** when a new event arrives, based on its type and arguments. The adapter is now defined as an **IIPCAdapter** interface, which is owned by the Arduino component. Events are exchanged through this interface.

The interface is defined in the **IPCAdapter**, and is implemented by the **TcpAdapter** and **SharedMemoryAdapter** projects. It declares methods for connecting to IPC and sending messages. It also declares a C# event called **ReceivedEventEvent**, which is invoked after decoding a message incoming from the backend, with the decoded **Event** struct as an argument. The Arduino component is subscribed to **ReceivedEventEvent**.

The **IPCAdapter** project also contains message encoders and a definition of the **Event** struct. The C++ and the C# version of the **Event** struct have the same signature, with the exception of owning a lambda function (described in 2.3.2). The lambda is not necessary for the frontend, as there will be only one source of events executed in the frontend, the ones received by the backend. They don't have to be put in a queue, they can be consumed immediately as they come. Frontend also produces events, e.g. when value on an Arduino's input pin is changed from the circuit, but those are never consumed by the frontend. It is the responsibility of the backend to handle them.

IEncoder and its **TextEncoder** implementation are analogous to **encoder.hpp** and **textEncoder.cpp** in the backend. The interface declares **EncodeEvent** and **TryDecodeEvent**, which transcribe events structs into human-readable strings and vice versa.

2.4.4 Utilities

Logger

The **Logger** project is analogous to its backend counterpart (2.3.3). It provides logging to a file with 4 levels of verbosity, that can be set up with a command line argument. Two loggers are used in the frontend as well, one for IPC and one for circuitry logs. There is one additional implementation of the **ILogger** called **CompositeLogger**, which holds list of loggers, and logs into all of them simultaneously. This mechanism anticipates a need for another logger implementation, which will display the log messages in the GUI using Avalonia.

3 Backend implementation

3.1 Simulator module

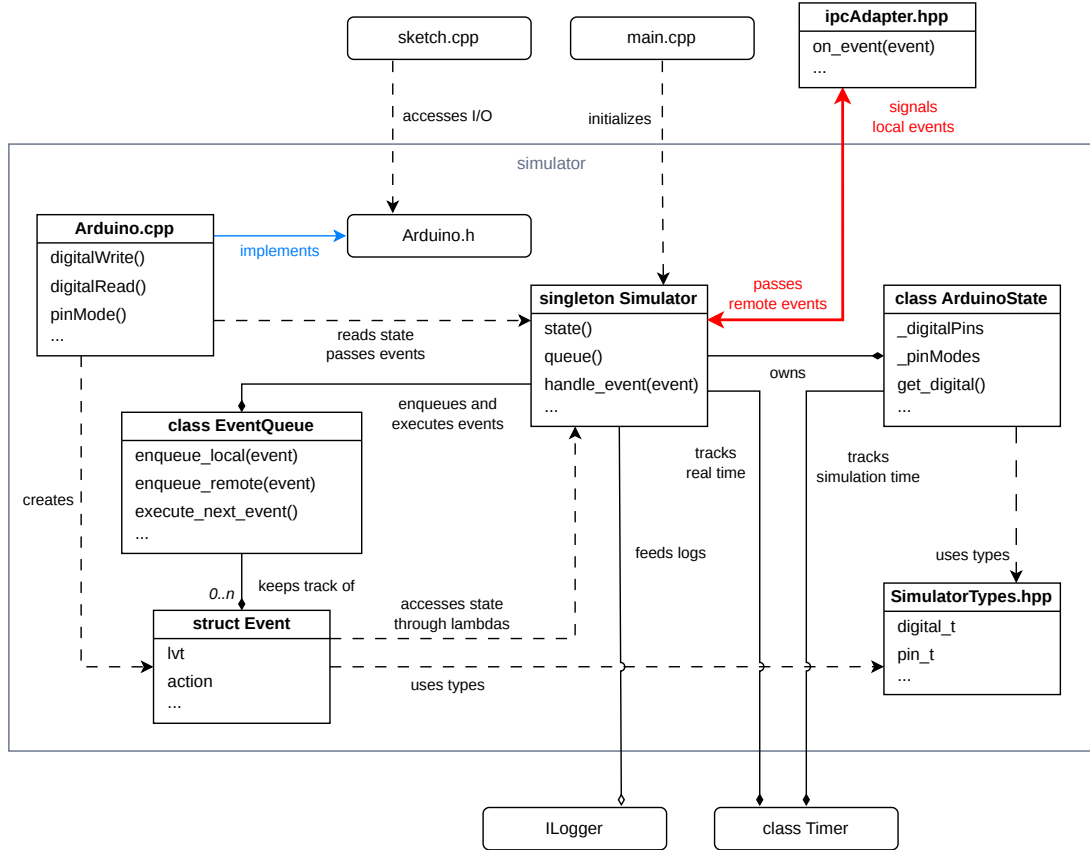


Figure 3.1 simulator module architecture

The `Simulator` class is initialized by the `main.cpp`, and its pointer is stored as a static field of the class itself, making it a singleton. It owns both the `ArduinoState` and `EventQueue` objects and provides access to them through its static classes. Namely `handle_event` is called by the `Arduino.cpp`, passing events to the queue and consuming all events that are already waiting.

The events are created with static factory classes of the `Event` struct. In these, lambda functions are being created, utilizing the `Simulator`'s `state()` method, which provides direct access to the `ArduinoState`.

3.1.1 ArduinoState class

Listing 3.1 ArduinoState class

```

1 class ArduinoState {
2     public:
3         static constexpr uint8_t ANALOG_PIN_COUNT = 6;
4         static constexpr uint8_t DIGITAL_PIN_COUNT = 14;
5         static constexpr uint8_t PIN_COUNT =
6             ANALOG_PIN_COUNT + DIGITAL_PIN_COUNT;

```

```

7
8     using analog_arr_t = std::array<analog_t, ANALOG_PIN_COUNT>;
9     using digital_arr_t = std::array<digital_t, DIGITAL_PIN_COUNT>;
10    using pinmode_arr_t = std::array<PinMode, PIN_COUNT>;
11
12    void init_timer();
13    float get_time() { return _timer.deltaTime(); }
14
15    analog_arr_t& get_analog() { return _analogPins; }
16    double get_analog(pin_t pin) const;
17    bool set_analog(pin_t pin, analog_t value);
18
19    digital_arr_t& get_digital() { return _digitalPins; }
20    bool get_digital(pin_t pin) const;
21    bool set_digital(pin_t pin, digital_t value);
22
23    pinmode_arr_t& get_pin_mode() { return _pinModes; }
24    PinMode get_pin_mode(pin_t pin) const;
25    bool set_pin_mode(pin_t pin, PinMode mode);
26 private:
27     Timer _timer;
28     analog_arr_t _analogPins = {};
29     digital_arr_t _digitalPins = {};
30     pinmode_arr_t _pinModes = {};
31 };

```

The class holds the digital states and modes of pins in `std::arrays`. Sizes of these arrays match the specification of the Arduino UNO R3 SMD board. The analog pins also stored in this class are not utilized at the point of writing this thesis. The pins can be read and modified with setters and getters analagous to those in `Arduino.h`, but instead of integers, the paramters are of types defined in `SimulatorTypes.hpp`. Namely `digital_t` is `bool`, `analog_t` is `double`, `pin_t` is `uint8_t` and `PinMode` is an enum attaining values `In` and `Out`.

The parameterless overload return reference to an entire array. This was particularly useful when processing snapshots, before events were implemented. The setter methods return `bool` indicating success of the operation. Currently all methods always indicate success, but to achieve higher quality of life for users, warnings and errors should be logged instead of crashing the simulator immediately, e.g. when accessing pins outside of array's range.

The `_timer` owned by this class represents the simulation time. The `Timer` class is not described in detail in this thesis as it was originally designed for a different project, and a lot of its features are not interesing for this particular work. Under the hood it uses `std::chrono` library. The `deltaTime()` method returns microseconds passed from the timer's last step. The timers of this thesis are stepped exclusively at its intialization.

3.1.2 Simulator singleton class

Listing 3.2 Simulator class

```

1 class Simulator {
2     public:
3         using event_callback_fnct =
4             std::function<void(const Event&)>;

```

```

5     using logger_ptr = std::shared_ptr<logger::ILogger>;
6
7     static void init(
8         event_callback_fnct eventCallback,
9         logger_ptr logger = nullptr);
10
11    static ArduinoState& state() { return s_instance->_state; }
12    static EventQueue& queue() { return s_instance->_eventQueue; }
13
14    static int64_t simulation_time() { return state().get_time(); }
15    static int64_t real_time()
16        { return s_instance->_timer.deltaTime(); }
17
18    static void handle_event(Event event);
19    static void handle_events();
20
21    static void log_debug(std::string&& str)
22        { s_instance->_logger->log_debug(std::move(str)); }
23    static void log_info(std::string&& str) { ... }
24    static void log_warning(std::string&& str) { ... }
25    static void log_error(std::string&& str) { ... }
26
27    private:
28        static Simulator* s_instance;
29        event_callback_fnct f_eventCallback;
30        logger_ptr _logger;
31        ArduinoState _state;
32        EventQueue _eventQueue;
33        Timer _timer;
34    };

```

The singleton's instance is created on the heap within the static `init` method and its pointer stored in the `s_instance` member. The `eventCallback` parameter of this method must be set to the handler, that should be called upon event consumption. In our implementation that is the `on_event` function from `ipcAdapter.hpp`. This callback is stored into `f_eventCallback`.

`state` and `queue` methods are used to access the objects owned by the `Simulator`. But the queue is more typically accessed through the `handle_event` and `handle_events` calls. They both consume all remote events waiting in the queue. The `handle_event` variant also puts a local event in the queue and consumes after all remote events with lower LVT.

The `Simulator` also owns a `Timer`, this one representing the real time passed. At the moment, both timers show roughly the same time, but it is anticipated, that the simulation timer could be manipulated by the user in the future versions of the project. Lastly, the `Simulator` class holds a shared pointer to a logger. It also exposes shortcut methods for passing log messages of different verbosity to the logger. This particular logger is used for any messages unrelated to the IPC.

handle_event method

Listing 3.3 `Simulator::handle_event` method

```

1 void Simulator::handle_event(Event event)
2 {
3     auto& queue = s_instance->_eventQueue;

```

```

4     queue.enqueue_local(event);
5
6     do {
7         queue.execute_next_event();
8     } while (event.id() != queue.last_executed_event().id());
9
10    s_instance->f_eventCallback(event);
11 }

```

The event is enqueued to the local event queue, and all remote events with lower LVT are consumed. To check that the local event has been consumed, we had to add a unique identifier member to the `Event` struct distinguishing different event instances. This `id` is compared to the `id` of the previously consumed event, and the while loop is stopped when these two match. We couldn't use the LVT value as, even though it is unique for local events, it doesn't have to be unique across processes. In some frontend implementations, remote events can actually have higher LVT than the local ones.

Due to the semantics of LVT, we ensure that every remote event that occurred before the local event has been consumed. The exception are events that already occurred remotely, but haven't been fast enough to reach the backend's event queue. This can occur with TCP connection, or with very frequent `Arduino.h` calls in the sketch. These events will most likely be still consumed in the next call of `handle_event`. But for full correctness more synchronization techniques should be implemented, depending on the use case.

When the local event has finally been consumed, it is signalled to the IPC through the `f_eventCallback`.

3.1.3 Event struct

Listing 3.4 Event struct

```

1 struct Event {
2     public:
3         enum class Type { Write, PinMode, Reboot };
4
5         Type type;
6         int64_t timestampMicros;
7         int64_t localVirtualTime;
8         std::function<void()> action;
9
10        static constexpr int MAX_ARGS = 3;
11        std::array<int, MAX_ARGS> args;
12
13        template <typename... Args>
14        Event(Type type, Args... arguments);
15        Event() {}
16
17        size_t id() const { return _id; }
18        std::string to_string();
19
20        static Event write(pin_t pin, digital_t value);
21        static Event set_pinmode(pin_t pin, PinMode mode);
22        static Event reboot();
23    private:

```

```

24     static size_t s_nextEventId;
25     size_t _id;
26 };

```

The members of the `Event` struct mostly match the event specification in the subsection 2.2.3. The additions are the `id` and `action` members. We already justified the use `id` member in the previous section. It is assigned in the constructor from the static `s_nextEventId`, which is then incremented. The `action` is assigned as a lambda function from within the static factory methods. This lambda is executed during the `EventQueue`'s `execute_next_event` call.

Write event

Listing 3.5 Write event factory

```

1  Event Event::write(pin_t pin, digital_t value)
2  {
3      Event event(Type::Write, pin, value);
4      event.action = [=]() {
5          Simulator::state().set_digital(pin, value);
6          Simulator::log_info(std::format(
7              "Written value {} to pin {}", value, pin));
8      };
9      return event;
10 }

```

The above is an example on how the factory methods should look like. The constructor takes a type and variadic arguments corresponding to that type. the constructor is statically asserted to not accept more than 3 variadic arguments. The action is defined as a lambda, with call to the `Simulator`'s `state` getter, modifying its values. The outcome is then logged through the `Simulator`'s logger.

At this point, the LVT of the event is not known. That is the responsibility of the scheduling entity. For local events it is the `EventQueue`, for remote events it is IPC, which has to decode LVT from an incoming message.

3.1.4 EventQueue class

Listing 3.6 EventQueue class

```

1  class EventQueue {
2  public:
3      void enqueue_local(Event& event);
4      void enqueue_remote(Event event);
5      void execute_next_event();
6
7      Event& last_executed_event()
8      { return _processedEvents.back(); }
9      bool is_empty() const
10     { return _localEvents.empty() && _remoteEvents.empty(); }
11     int64_t next_lvt() const { return _nextLvt; }
12     int64_t last_lvt() const { return _lastLvt; }
13
14 private:
15     std::queue<Event> _localEvents;
16     std::queue<Event> _remoteEvents;

```

```

17     std::queue<Event> _processedEvents;
18     int64_t _nextLvt;
19     int64_t _lastLvt;
20
21     void clear();
22     size_t earliest_queue_lvt(const std::queue<Event>& queue);
23 };

```

The event queue is in reality two queues **lie**. One storing local and one storing remote events. As we only have two threads running concurrently in the backend, this simplifies the synchronization significantly. We also completely avoid locking the structure in order to prevent race conditions. These queues are populated with the public `enqueue_local` and `enqueue_remote` methods.

nextLvt, processedEvents

execute_next_event method

Listing 3.7 execute_next_event method

```

1 void EventQueue::execute_next_event()
2 {
3     auto& queueAhead =
4         (earliest_queue_lvt(_localEvents) <
5          earliest_queue_lvt(_remoteEvents))
6         ? _localEvents
7         : _remoteEvents;
8
9     Event processedEvent = queueAhead.front();
10    queueAhead.pop();
11
12    processedEvent.action();
13    _nextLvt = std::max(
14        _nextLvt, processedEvent.localVirtualTime + 1);
15    _processedEvents.push(processedEvent);
16 }

```

Whichever event that has the minimal LVT of both queues has to be executed next. That is to preserve the semantics of LVT, such that every event with higher LVT is the consequence of an event with lower LVT. We expect that the queues are sorted by the LVT of their events, from lowest to highest. For local events that is given by sequential execution of the sketch. For remote events this is given by the IPC technologies we've chosen. TCP guarantees that data are received in the same order as they were sent. In the shared memory approach, we deliberately write and store messages in a sequential order. So to determine the next event, all we have to do is peek at the first element of both of the queues, and select the one with lower LVT. That is ultimately how `queueAhead` is obtained.

The event is then popped from the queue. The `action` lambda stored within it is finally executed. The event can now be considered consumed. Then the `_nextLvt` is decided. If the consumed event has the highest encountered LVT yet, we simply increment this value. If it does not, we don't change the value, as it would mean the subsequent events would technically occur in the past.

3.1.5 Arduino.cpp

Listing 3.8 Arduino.cpp

```

1 using namespace arguino::simulator;
2
3 void digitalWrite(uint8_t pin, uint8_t val)
4 {
5     if (Simulator::state().get_digital(pin) == val) return;
6
7     Simulator::handle_event(
8         Event::write(pin, val == HIGH)
9     );
10 }
11
12 int digitalRead(uint8_t pin)
13 {
14     Simulator::handle_events();
15     return Simulator::state().get_digital(pin);
16 }
17
18 unsigned long millis()
19 {
20     return std::floor(Simulator::state().get_time() / 1000);
21 }
22
23 void pinMode(uint8_t pin, uint8_t mode)
24 {
25     PinMode pinMode = mode == OUTPUT ? PinMode::Out : PinMode::In;
26     if (Simulator::state().get_pin_mode(pin) == pinMode) return;
27
28     Simulator::handle_event(
29         Event::set_pinmode(pin, pinMode)
30     );
31 }

```

At last, we implement the four functions specified in the goals of the thesis. `millis` simply takes the simulation time and divides it by 1000 to get the elapsed milliseconds. The `digitalWrite` and `pinMode` functions are implemented using the `handle_event` call with the corresponding event factory passed as an argument. If the argument for these events doesn't differ from the current value stored in `ArduinoState`, we exit the function early. In sketches with these functions in the outermost loop scope, which is actually a common pattern, the condition can save millions of redundant events and log messages.

The `digitalRead` call consumes all waiting events, and then simply reads the pin of `ArduinoState`. Out of the four functions, this is the only one that can be problematic with regard to delayed event reception described in the `handle_event` method (3.1.2), as it is the only one dependant on previous `Arduino.h` calls. It is possible for `digitalRead` to return a state outdated by a few microseconds.

A proposed solution is to add an event serving as a kind of condition variable. This event would be sent to the frontend and the same event would be sent back after any necessary frontend changes would have been resolved. Until then, the backend's simulation thread would remain in a spin lock, pausing the simulation timer, handling remote events and checking for the expected synchronization event. Only then the function would return a value and the execution of the sketch would resume.

This proposal haven't been implemented at the time of writing the thesis

due to time concerns. This incorrect behavior hasn't been yet observed with the provided tests, and it probably won't be noticed on simpler sketches of a target user audience.

3.2 Executable module

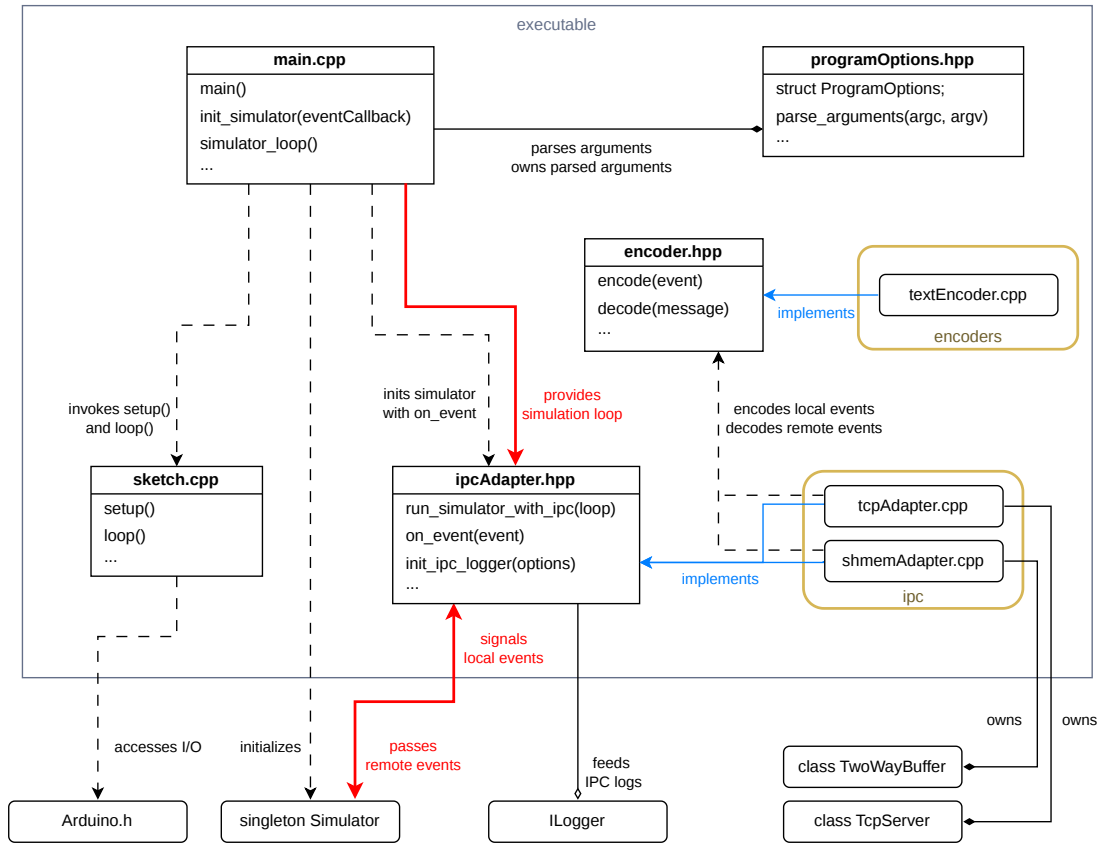


Figure 3.2 executable module architecture

The `executable` module puts all the systems together and compiles into an executable file. `sketch.cpp` doesn't come with the source code. It has to be preprocessed from the sketch user wishes to simulate. `main.cpp` defines how the sketch is used, and orders the IPC adapter to use it in context of the used IPC mechanism.

Two execution targets are provided in the CMake configuration, one for each IPC mechanism. They differ in which IPC adapter implementation is used, and which program options are supported with that implementation. They also differ in whether the `tcp-server` or the `shared-memory` module is linked with the executable.

3.2.1 main.cpp

Listing 3.9 main.cpp

```
1 using logger_t = logger::FileLogger;
2 using namespace arguino::simulator;
```

```

3 ProgramOptions options;
4
5 void init_simulator(
6     Simulator::event_callback_fnct eventCallback)
7 {
8     auto logPath = std::filesystem::absolute(
9         options.SimulatorLogPath);
10    auto logger = std::make_shared<logger_t>(logPath);
11
12    Simulator::init(eventCallback, logger);
13    logger->timerFunction = Simulator::simulation_time;
14    logger->verbosityLevel = options.Verbosity;
15 }
16
17 void simulator_loop()
18 {
19     Simulator::state().init_timer();
20
21     Simulator::handle_event(Event::reboot());
22     Simulator::log_info("Simulator initialized.");
23     setup();
24     Simulator::log_info(
25         "setup() finished successfully. Running loop().");
26
27     while (true) {
28         loop();
29     }
30 }
31
32 int main(int argc, char** argv)
33 {
34     options = parse_arguments(argc, argv);
35     init_simulator(on_event);
36     run_simulator_with_ipc(simulator_loop, options);
37 }

```

The `main` function of the C++ project parses the command line arguments, initializes the Simulator singleton, defines the simulation loop and let's the IPC adapter handle the rest. The `on_event` function passed to `init_simulator` is included from `ipcAdapter.hpp` and implemented by the adapter chosen via CMake execution target. Simulation logger and it's verbosity are created in the `init_simulator` as well.

The `simulator_loop` calls the the required functions included from `sketch.cpp`: the `setup` function once and the `loop` function in an infinite while loop. The *Reboot* event is sent as the very first event of each simulation. This is useful when the frontend keeps running, but backend is being restarted or recompiled with a different sketch. The lack of such an event would preserve the frontend state of the old simulation, and it would get desynchronized very quickly.

The command line arguments are parsed using the `boost::program_options` library. The values in their expected data types are stored in the `ProgramOptions` struct. An instance of this struct is stored statically in the `main.cpp` file, and passed down to the IPC adapter by reference.

3.2.2 ipcAdapter.hpp

Listing 3.10 ipc_adapter.hpp

```

1 using logger_t = logger::FileLogger;
2 using loop_fnct = std::function<void()>;
3 using init_fnct = std::function<
4     void(arguino::simulator::Simulator::event_callback_fnct)>;
5
6 static std::shared_ptr<logger_t> IpcLogger;
7
8 void run_simulator_with_ipc(
9     loop_fnct loopFunction,
10     const ProgramOptions& options);
11
12 void on_event(const arguino::simulator::Event& event);
13
14 inline void init_ipc_logger(const ProgramOptions& options)
15 {
16     auto loggerAbsolutePath = std::filesystem::absolute(
17         options.IcpLogPath);
18     IpcLogger = std::make_shared<logger_t>(loggerAbsolutePath);
19
20     IpcLogger->verbosityLevel = options.Verbosity;
21     IpcLogger->timerFunction =
22         arguino::simulator::Simulator::real_time;
23 }

```

The responsibility of the `run_simulator_with_ipc` method is to initialize the classes used for IPC and store them in the implementing `.cpp` file. That typically involves creating a new thread for listening to messages incoming from the frontend process. This thread has to be joined at some point after the simulation loop. That is why we pass the loop function as an argument, to be invoked at the right time.

The `on_event` function is passed to the `Simulator` and is called every time a local event is consumed. The event is encoded into a message using the function included from `encoder.hpp`. The message can be then transmitted over the IPC, which has already been initialized before the very first local event is created.

The `init_ipc_logger` function is already defined as it provides the IPC logging setup common for any adapter. We call this function at the beginning of each `run_simulator_with_ipc` definition.

Implementations are located in the `ipc` directory of the `executable` module. But we should first understand how were the IPC mechanisms implemented, we moved the adapter documentations to their respective chapters (subsections 3.3.3 and 3.4.4)

3.2.3 textEncoder.cpp

Listing 3.11 encoder.hpp

```

1 constexpr char UNKNOWN_EVENT[] = "UNKNOWN";
2
3 std::string encode_event(
4     const arguino::simulator::Event& event);
5 bool decode_event(
6     const std::string& message,
7     arguino::simulator::Event& event);

```

The `encoder.hpp` declares a function that encodes local events into messages, and decodes messages into remote events `decode_event` outputs the result event as a reference parameter. Its return value indicates success of the operation. That is, whether or not was the received message in a valid a format. The `UNKNOWN_EVENT` constant is used to identify undefined event types.¹

Listing 3.12 `textEncoder.cpp`

```

1 using namespace arguino::simulator;
2
3 constexpr char PART_DELIMITER = ':';
4 constexpr char COMMON_PART_FORMAT[] = "{:012}:{:07}:";
5 constexpr char WRITE_FORMAT[] = "W:{:02}:{:02}";
6 constexpr char PINMODE_FORMAT[] = "P:{:02}:{:02}";
7 constexpr char REBOOT_FORMAT[] = "R";
8
9 constexpr int COMMON_PARTS = 3;
10 constexpr int MAX_ARGS = 3;
11 constexpr int MAX_PARTS = 6;
12
13 using parts_array_t = std::array<std::string_view, MAX_PARTS>;
14
15 static std::string encode_write(const Event& e)
16     { return std::format(WRITE_FORMAT, e.args[0], e.args[1]); }
17
18 static std::string encode_pinmode(const Event& e)
19     { return std::format(PINMODE_FORMAT, e.args[0], e.args[1]); }
20
21 static std::string encode_reboot(const Event& e)
22     { return REBOOT_FORMAT; }
23
24 std::string encode_event(const Event& event)
25 {
26     std::string commonPart = std::format(
27         COMMON_PART_FORMAT,
28         event.timestampMicros,
29         event.localVirtualTime);
30
31     switch (event.type) {
32         case Event::Type::Write:
33             return commonPart + encode_write(event);
34         case Event::Type::PinMode:
35             return commonPart + encode_pinmode(event);
36         case Event::Type::Reboot:
37             return commonPart + encode_reboot(event);
38         default:
39             return UNKNOWN_EVENT;
40     }
41 }
42
43 bool decode_int(std::string_view str, int64_t& out) { ... }
44 bool decode_int(std::string_view str, int& out) { ... }
45 parts_array_t split(std::string_view messageView) { ... }
46
47 bool decode_event(const std::string& message, Event& event)

```

¹This can occur when implementing new events, but forgetting to add new cases in encoding. This should never happen with the source code attached with this thesis.

```

48 {
49     parts_array_t parts = split(message);
50     std::array<int, MAX_ARGS> args;
51
52     for (int i = 0; i < MAX_ARGS; ++i) {
53         if (!decode_int(parts[COMMON_PARTS + i], args[i])) break;
54     }
55
56     if (parts[2] == "W") {
57         event = Event::write(args[0], args[1]);
58     }
59     else if (parts[2] == "P") {
60         event = Event::set_pinmode(args[0], (PinMode)args[1]);
61     }
62     else {
63         return false;
64     }
65     if (!decode_int(parts[0], event.timestampMicros) ||
66         !decode_int(parts[1], event.localVirtualTime)) {
67         return false;
68     }
69     return true;
70 }

```

At the time of writing this thesis, `textEncoder.cpp` is the only encoder implementation. It handles human readable messages, the format of which is described in more detail in the section 5.3.1. The message is divided into 3 to 6 parts delimited by a colon, the first 2 of which are common for every event type. Hence the constant string formats for each of the types. `encode_message` identifies the type, fills the event values into the respective format, and its done.

`decode_message` has to manually split the message into substrings and parse each substring individually. First we decode arguments in a loop. then we decode the type encoding character, and based on that we call the appropriate factory method with the right arguments. In the end we decode the timestamps from the common parts.

At each of these parts we can encounter a value that doesn't conform to its expected data type or a known event type, and the decoding fails. Then we have to return `false`. At this point, the event can be already partially initialized, but it is the responsibility of the caller to discard this instance, and fall into some fail state.

3.3 TCP Server module

3.3.1 TcpServer class

Listing 3.13 TcpServer class

```

1 class TcpServer {
2     public:
3         using handler_t = ConnectionHandler;
4         using logger_ptr = std::shared_ptr<logger::ILogger>;
5
6         TcpServer(
7             uint16_t port,

```

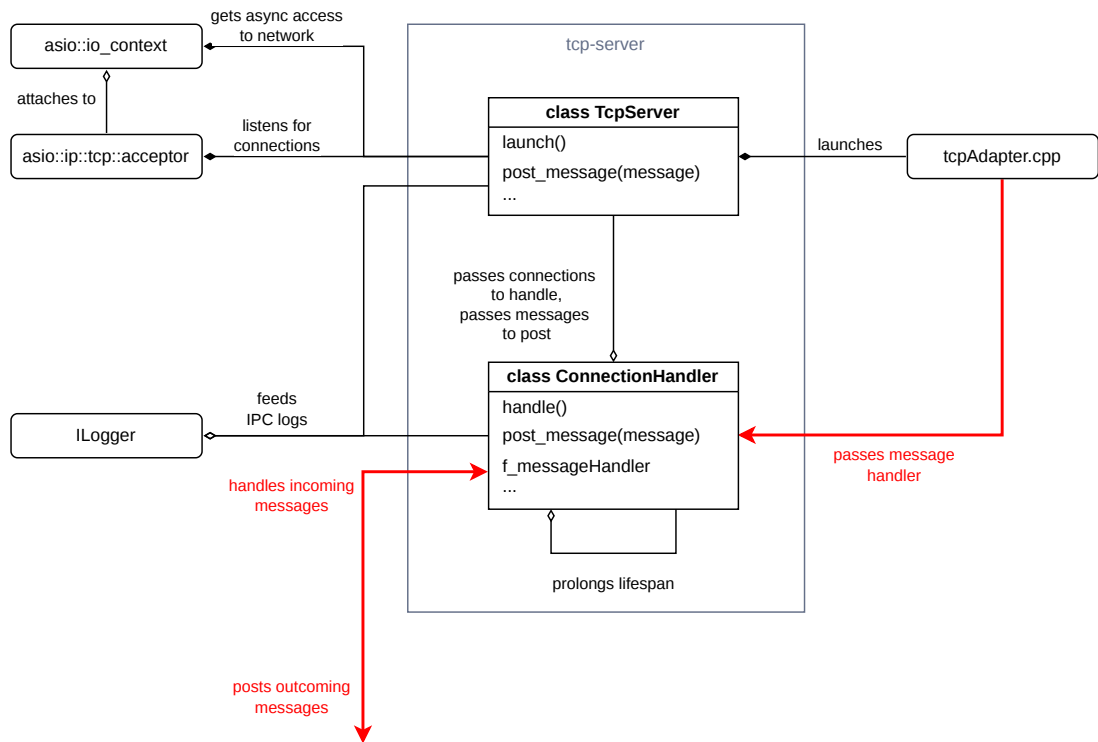


Figure 3.3 tcp-server module architecture

```

8         handler_t::message_funcnt messageHandler,
9         logger_ptr logger);
10
11 void launch();
12 void post_message(const std::string& message);
13 bool is_connected() const;
14
15 private:
16     boost::asio::io_context _ioContext;
17     boost::asio::ip::tcp::endpoint _endpoint;
18     boost::asio::ip::tcp::acceptor _acceptor;
19     uint16_t _port;
20
21     std::shared_ptr<handler_t> _connectionHandler;
22     std::mutex _connectionMutex;
23     std::condition_variable _connectionCv;
24     handler_t::message_funcnt _messageHandler;
25     logger_ptr _logger;
26
27     void start_accepting();
28 };

```

3.3.2 ConnectionHandler class

Listing 3.14 ConnectionHandler class

```

1 class ConnectionHandler :
2     public std::enable_shared_from_this<ConnectionHandler> {
3 public:
4     static constexpr char MESSAGE_DELIMITER = ',';

```

```

5 static constexpr size_t MAX_QUEUE_SIZE = 100;
6 using self_ptr = std::shared_ptr<ConnectionHandler>;
7 using logger_ptr = std::shared_ptr<logger::ILogger>;
8 using message_funct = std::function<void(const std::string&)>;
9
10 ConnectionHandler(
11     boost::asio::io_context& ioContext,
12     message_funct messageHandler,
13     logger_ptr logger);
14 static self_ptr create(
15     boost::asio::io_context& ioContext,
16     message_funct messageHandler,
17     logger_ptr logger);
18
19 void handle();
20 void post_message(const std::string& message);
21 boost::asio::ip::tcp::socket& socket()
22     { return _socket; }
23 bool is_connected() const { return _isConnected; }
24
25 private:
26     boost::asio::io_context::executor_type _executor;
27     boost::asio::ip::tcp::socket _socket;
28     std::string _incomingBuffer;
29     std::queue<std::string> _outboxQueue;
30     std::mutex _queueMutex;
31     std::condition_variable _queueCv;
32     size_t _connectionId;
33     static size_t _nextConnectionId;
34     bool _isConnected;
35     std::atomic<bool> _isWriting;
36     message_funct _messageHandler;
37     logger_ptr _logger;
38
39     void handle_message(
40         boost::system::error_code error,
41         size_t messageSize);
42     void write_from_queue();
43     void disconnect();
44 };

```

3.3.3 tcpAdapter.cpp

Listing 3.15 tcp_adapter.cpp

```

1 using namespace arguino::simulator;
2 using tcp_server_t = arguino::tcp::TcpServer;
3 static std::unique_ptr<tcp_server_t> _tcpServer;
4
5 void on_event(const Event& event)
6 {
7     auto message = encode_event(event);
8     if (message == UNKNOWN_EVENT) {
9         IpcLogger->log_error(
10             "Encountered unknown event type; skipping.");
11         return;
12     }

```



```

13
14     IpLogger->log_debug("Enqueuing message " + message);
15     _tcpServer->post_message(message);
16 }
17
18 static void on_received_tcp_message(const std::string message)
19 {
20     Event event;
21     if (decode_event(message, event)) {
22         Simulator::queue().enqueue_remote(event);
23     }
24     else {
25         IpLogger->log_error(
26             "Could not decode event from message " + message);
27     }
28 }
29
30 static void run_tcp() { _tcpServer->launch(); }
31
32 void run_simulator_with_ipc(
33     loop_fnct loopFunction,
34     const ProgramOptions& options)
35 {
36     init_ipc_logger(options);
37     _tcpServer = std::make_unique<tcp_server_t>(
38         options.TcpPort, on_received_tcp_message, IpLogger);
39
40     std::thread tcp_thread(run_tcp);
41     IpLogger->log_info("Tcp Server launched");
42     loopFunction();
43     tcp_thread.join();
44 }

```

3.4 Shared Memory module

The `shmemAdapter.cpp` holds a `TwoWayBuffer` object. It represents the two circular buffers, one producing and one consuming. It creates the OS-level memory mapped object using the `boost::interprocess` library, and splits it into two regions of the same size, one for each buffer.

The `CircularBuffer` class implements the data structure of the buffer itself. It was designed deliberately for use with shared memory. It holds the `MemoryRegion` instance, which abstracts away the low level details of accessing the shared memory directly. We defined a forward iterator for the `CircularBuffer` to simplify the manipulation of its actual data.

3.4.1 MemoryRegion class

Listing 3.16 MemoryRegion class

```

1 class MemoryRegion {
2     public:
3         using shmem_t =
4             boost::interprocess::shared_memory_object;
5         using shmem_region_t =

```

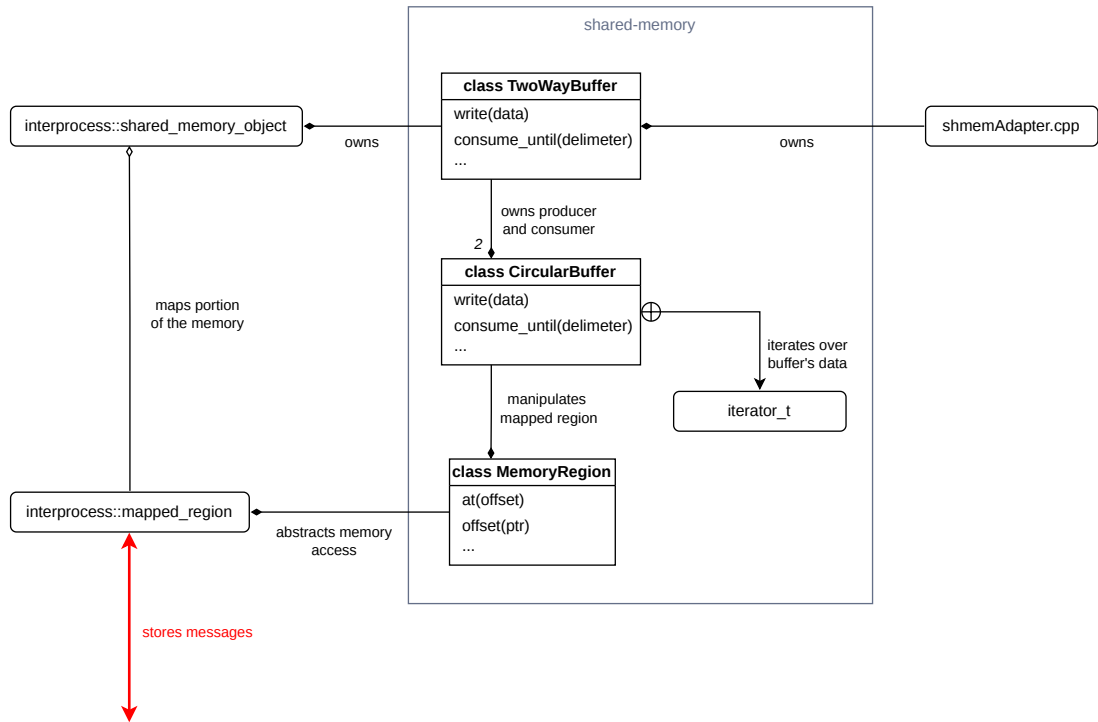


Figure 3.4 shared-memory module architecture

```

6         boost::interprocess::mapped_region;
7
8     MemoryRegion(
9         const shmem_t& shmemObject,
10        size_t offset,
11        size_t size);
12
13    size_t size() { return _mappedRegion.get_size(); }
14    uint8_t* begin()
15        { return (uint8_t*)_mappedRegion.get_address(); }
16    uint8_t* end() { return begin() + size(); }
17
18    template <typename T>
19    T& at(size_t offset);
20    uint8_t* at(size_t offset) { return begin() + offset; }
21    size_t offset(void* ptr)
22        { return (uint8_t*)ptr - begin(); }
23
24    private:
25        shmem_region_t _mappedRegion;
26 };

```

This class provides an abstraction over the `interprocess::mapped_region` type. The aforementioned type isolates a region of memory within the `interprocess::shared_memory_object` and provides direct access to it. We create the `mapped_region` object in the constructor, defined by its size and the offset from the starting address of the `shared_memory_object` in bytes. We set its permission to read and write.

The methods of `MemoryRegion` consist mostly of transformations from `void*` to `uint8_t*`, and conversions between pointers and offsets. We cannot perform

pointer arithmetic on `void*`, so it is useful to have the pointers already in form that allows passing through memory byte by byte. We already mentioned the importance of offsets in the section 2.2.5.

The templated `at` method uses reinterpretation casting to automatically retrieve the underlying type from raw bytes, and returns a reference to.

3.4.2 CircularBuffer class

Listing 3.17 CircularBuffer class

```

1 class CircularBuffer {
2     public:
3         struct iterator_t;
4         friend struct iterator_t;
5         using shmem_t =
6             boost::interprocess::shared_memory_object;
7
8         static constexpr size_t PRODUCER_PTR_LOCATION = 0;
9         static constexpr size_t CONSUMER_PTR_LOCATION = 8;
10        static constexpr size_t BUFFER_LOCATION = 16;
11        CircularBuffer() = default;
12        CircularBuffer(
13            const shmem_t& shmemObject,
14            size_t offset,
15            size_t size);
16
17        bool is_empty();
18        size_t buffer_size() { ... }
19        size_t bytes_filled();
20        size_t bytes_available() { ... }
21
22        template <typename T>
23            requires std::is_trivially_copyable_v<T>
24                && (!std::ranges::contiguous_range<T>)
25        bool write(const T& data);
26
27        template <std::ranges::contiguous_range TRange>
28        bool write(const TRange& data);
29
30        std::vector<uint8_t> consume_until(uint8_t delimiter);
31
32    private:
33        std::unique_ptr<MemoryRegion> _memoryRegion;
34
35        iterator_t begin();
36        iterator_t end();
37        iterator_t at(size_t offset);
38        iterator_t producer_it();
39        iterator_t consumer_it();
40        void write_producer(iterator_t producer_it);
41        void write_consumer(iterator_t consumer_it);
42 };

```

The data structure occupies the entire space defined by the underlying `MappedRegion`. The producer and consumer offsets occupy the first 16 bytes of the region. The rest is filled with the raw data, simply referred to as `buffer` in

the code. In the context of this structure, the offset must be interpreted as the distance from the address from the start of the buffer.

To simplify this notion we defined a forward iterator with this offset as the underlying iteration unit. It dereferences into `uint8_t`. Incrementing over the boundary of the memory region sets the offset back to 0. `producer_it` and `consumer_it` turn the offsets stored in the raw memory into these iterators. They can be written back with the `write_producer` and `write_consumer` methods.

For the purpose of writing and reading messages (or bigger structures in general), we define the `write` and `consume_until` methods. These implicitly access the data at `producer_it`, respectively `consumer_it`, onwards. `write` comes in two variants, one for contiguous ranges and one for any other templated type. The return value indicates whether or not does the buffer have enough available space to insert the type. `consume_until` takes a delimiter as the point at which to stop reading the data. It returns the read bytes in an `std::vector` and moves the producer offset.

write method

Listing 3.18 `CircularBuffer::write` method

```
1 template <std::ranges::contiguous_range TRange>
2 inline bool CircularBuffer::write(const TRange& data)
3 {
4     auto bytes = std::as_bytes(std::span(data));
5
6     // We don't want the buffer to be fully filled
7     if (bytes_available() <= bytes.size()) {
8         return false;
9     }
10
11     size_t bytesUntilBufferEnd = end() - producer_it();
12
13     if (bytes.size() < bytesUntilBufferEnd) {
14         std::memcpy(
15             &*producer_it(), bytes.data(), bytes.size());
16     }
17     else { // Message written on a boundary
18         auto firstBytes = bytes.first(bytesUntilBufferEnd);
19         auto lastBytes = bytes.last(
20             bytes.size() - bytesUntilBufferEnd);
21
22         std::memcpy(
23             &*producer_it(), firstBytes.data(), firstBytes.size());
24         std::memcpy(
25             &*begin(), lastBytes.data(), lastBytes.size());
26     }
27
28     write_producer(producer_it() + bytes.size());
29     return true;
30 }
```

The range variant of `write` is the one that does all the hard work. The other one just transforms the type `T` into an `std::span` with the constant reference to `T` as its only element. This qualifies it as a range.

First we turn the range into a span of bytes, so we can access each one of them using pointer arithmetics. Then we check if there is enough available space to store all the bytes. We have to make sure that there is always at least one byte unused. Otherwise would the consumer and producer offsets equal. We specifically interpret this state as the buffer being empty. Without this specification it would be ambiguous whether is the buffer empty or full.

Then we check if there is enough space left until the end of the buffer. If not, we will have to write the data on its boundary by splitting it in two. The `firstBytes` get copied to the producer offset, and the rest at the offset 0.

consume_until method

Listing 3.19 `CircularBuffer::consume_until` method

```

1  std::vector<uint8_t> CircularBuffer::consume_until(
2      uint8_t delimiter)
3  {
4      iterator_t producerIt = producer_it();
5      iterator_t consumerIt = consumer_it();
6
7      auto delimiterIt = std::find(
8          consumerIt, producerIt, delimiter);
9      bool delimiterNotFound = delimiterIt == producerIt;
10     if (delimiterNotFound) return {};
11
12     // +1 to include the delimiter
13     size_t dataLength = delimiterIt - consumerIt + 1;
14     size_t wrapsAround = dataLength > end() - consumerIt;
15     std::vector<uint8_t> result(dataLength);
16
17     if (!wrapsAround) {
18         std::memcpy(&result[0], &*consumerIt, dataLength);
19     }
20     else { // message lies on the boundary
21         size_t firstBytesLength = end() - consumerIt;
22         size_t lastBytesLength = delimiterIt - begin();
23
24         std::memcpy(&result[0], &*consumerIt, firstBytesLength);
25         std::memcpy(
26             &result[firstBytesLength], &*begin(), lastBytesLength);
27     }
28
29     write_consumer(delimiterIt + 1);
30     return result;
31 }
```

First we need to copy the producer offset as it can change during the execution of the function. Since `iterator_t` is a fully qualified forward iterator, we can pass the offset iterators as arguments of `std::find` in order to find the iterator of the delimiter. If the delimiter hasn't been found, we return an empty vector, indicating that the operation doesn't make sense with the current state of the buffer.

Otherwise we proceed analogously to `write`. We determine the length of the message, and if it could fit in until the end of the buffer. If not, it lies on the

boundary, so we have to copy portion of the data from the end of the buffer, and portion from its beginning.

3.4.3 TwoWayBuffer class

Listing 3.20 TwoWayBuffer class

```
1 class TwoWayBuffer {
2     public:
3         using shmem_t =
4             boost::interprocess::shared_memory_object;
5
6         TwoWayBuffer(const std::string& shmemName, size_t sizeInPages);
7         ~TwoWayBuffer();
8
9         template <typename T>
10        bool write(const T& data);
11
12        std::vector<uint8_t> consume_until(uint8_t delimiter);
13        CircularBuffer& producer() { return _producer; }
14        CircularBuffer& consumer() { return _consumer; }
15        std::string& name() { return _name; }
16        size_t pages() { return _pages; }
17        size_t size();
18
19    private:
20        std::string _name;
21        size_t _pages;
22        shmem_t _shmemObject;
23        CircularBuffer _producer;
24        CircularBuffer _consumer;
25 };
```

TwoWayBuffer is a wrapper around the CircularBuffer class. It owns the instances of both buffers, and limits access to them. `write` passes its arguments to `write` method of the producer buffer and `consume_until` passes its argument to the method of the same name of the consumer buffer.

The class also creates and owns the `shared_memory_object`. This object either creates a new OS-level shared memory object, or opens one with the given name if it already exists. Opening the object means it doesn't need to be destroyed, and the frontend doesn't need to shift into a panic state. At the construction of the circular buffers, both of the offsets are reset 0, effectively clearing any previous data.

The memory object needs to be specified its size, which is always a multiple of the page size on the given OS (usually 4096B). TwoWayBuffer then dedicates half the size of the memory object to each of the circular buffers.

3.4.4 shmemAdapter.cpp

Listing 3.21 shmem_adapter.cpp

```
1 using namespace arguino::shmem;
2 using namespace arguino::simulator;
3
```

```

4 constexpr uint8_t MESSAGE_DELIMITER = ',';
5 static TwoWayBuffer* _shmem;
6
7 void on_event(const Event& event)
8 {
9     auto message = encode_event(event);
10    if (message == UNKNOWN_EVENT) {
11        IpcLogger->log_error("...");
12        return;
13    }
14    if (!_shmem->write(message + ',')) {
15        IpcLogger->log_error("...");
16    }
17    else {
18        IpcLogger->log_debug("...");
19    }
20 }
21
22 static void run_shmem_consumer()
23 {
24     while (true) {
25         while (!_shmem->consumer().is_empty()) {
26             std::vector<uint8_t> data = _shmem->consumer()
27                 .consume_until(
28                     MESSAGE_DELIMITER);
29
30             if (data.empty()) {
31                 IpcLogger->log_error("...");
32                 continue;
33             }
34
35             std::string message(data.begin(), data.end());
36             IpcLogger->log_debug("...");
37
38             Event event;
39             if (decode_event(message, event)) {
40                 Simulator::queue().enqueue_remote(event);
41             }
42             else {
43                 IpcLogger->log_error("...");
44             }
45         }
46     }
47 }
48
49 void run_simulator_with_ipc(
50     loop_fnct loopFunction,
51     const ProgramOptions& options)
52 {
53     init_ipc_logger(options);
54     _shmem = new TwoWayBuffer(
55         options.ShmemName, options.ShmemSizePages);
56
57     std::thread consumer_thread(run_shmem_consumer);
58     IpcLogger->log_info("Shared memory buffer initialized.");
59
60     loopFunction();
61     consumer_thread.join();

```

The adapter utilizes the shared memory through the high level `TwoWayBuffer` wrapper. It provides it its name and size in pages from options. Then it creates a single thread that infinitely checks for new messages in the consumer buffer. Instead of the spin lock (inner while loop in `run_shmem_consumer`) it would be more appropriate to use interprocess semaphore to check the contents of the buffer, saving some resources. Inside the loop a message delimited by a semicolon is retrieved, decoded and enqueued. `on_event` method is as simple as encoding the event and writing it to the buffer, given there is enough space.

4 Frontend implementation

4.1 Component Management project

4.1.1 Pin class

Listing 4.1 Pin class

```
1 public record struct PinPrototype(uint Id, string? Name = null);
2
3 public class Pin
4 {
5     public uint Id { get; init; }
6     public string? Name { get; init; }
7
8     public bool IsDriving { get; private set; }
9     public DigitalState State =>
10         (IsDriving || (_node?.IsHigh ?? false))
11         ? DigitalState.High
12         : DigitalState.Low;
13
14     public bool IsHigh => State == DigitalState.High;
15     public bool IsLow => State == DigitalState.Low;
16     public bool IsWriteOnly => _mode == Mode.WriteOnly;
17     public bool IsReadOnly => _mode == Mode.ReadOnly;
18     public bool IsConnected => _node != null;
19
20     public event Action<Pin, bool>? DrivingChangedEvent;
21     public event Action<Pin, DigitalState>? StateChangedEvent;
22     public event Action<Pin>? PinConnectedEvent;
23     public event Action<Pin>? PinDisconnectedEvent;
24
25     private Component _component;
26     private ElectricalNode? _node;
27
28     public enum Mode { General, ReadOnly, WriteOnly }
29     private Mode _mode = Mode.General;
30
31     public Pin(Component component, PinPrototype prototype)
32     { ... }
33
34     public Pin(Component component, uint id, string? name = null)
35     { ... }
36
37     public void MakeReadOnly() => _mode = Mode.ReadOnly;
38     public void MakeWriteOnly() => _mode = Mode.WriteOnly;
39     public void MakeGeneral() => _mode = Mode.General;
40
41     public void SetHigh() => SetValue(true);
42     public void SetLow() => SetValue(false);
43
44     public void SetValue(DigitalState value) =>
45         SetValue(value == DigitalState.High);
46
47     public void SetValue(bool value) { ... }
48     public void SetMode(Mode mode) { ... }
```

```

49
50     public void ConnectToNode(ElectricalNode node) { ... }
51     public void Disconnect() { ... }
52
53     public void NotifyStateChange(
54         ElectricalNode? _,
55         DigitalState nodeState) { ... }
56 }

```

4.1.2 ElectricalNode class

Listing 4.2 ElectricalNode class

```

1 public class ElectricalNode {
2     public uint DrivingPins { get; private set; } = 0;
3     public DigitalState State => DrivingPins > 0
4         ? DigitalState.High
5         : DigitalState.Low;
6
7     public bool IsHigh => State == DigitalState.High;
8     public bool IsLow => State == DigitalState.Low;
9
10    public event Action<ElectricalNode, DigitalState>?
11        StateChangedEvent;
12
13    private List<Pin> _pins = new();
14
15    public void AddPin(Pin pin) { ... }
16    public void RemovePin(Pin pin) { ... }
17    private void Propagate(Pin sourcePin, bool beganDriving)
18    { ... }
19 }

```

4.1.3 Component abstract class

Listing 4.3 Component class

```

1 public abstract class Component {
2     public required string Name { get; set; }
3     public string TypeName { get; init; }
4
5     public ComponentConfiguration Configuration =>
6         _configs[TypeName];
7     public Transform Transform { get; set; } = new();
8     public SKSvg CurrentSprite { get; internal set; } = null!;
9
10    public event Action<SKSvg>? SpriteChangedEvent;
11
12    private readonly List<Pin> _pins = [];
13    private static readonly Dictionary
14        <string, ComponentConfiguration> _configs = [];
15
16    internal Component(string typeName) {
17        TypeName = typeName;
18    }
19 }

```

```

20 public virtual void OnPinConnected(Pin pin) {}
21 public virtual void OnPinDisconnected(Pin pin) {}
22 public virtual void OnPinStateChanged(Pin pin) {}
23 public virtual void OnControlPress(Vector2 cursorPosition) {}
24 public virtual void OnControlRelease() {}
25 internal virtual void OnInitialized() {}
26
27 public Pin? GetPin(string name) { ... }
28 public Pin? GetPin(uint id) { ... }
29
30 internal static void AddConfiguration(
31     ComponentConfiguration config) { ... }
32
33 protected void UpdateSprite(SKSvg sprite) { ... }
34 protected void UpdateSprite(string spriteName) { ... }
35
36 internal void InitPins() { ... }
37
38 private void OnPinStateChanged(Pin pin, DigitalState _) =>
39     OnPinStateChanged(pin);
40 }

```

4.1.4 SceneFactory static class

Listing 4.4 SceneFactory class

```

1 public class SceneFactory {
2     public Scene Scene { get; private set; } = new();
3
4     private string _scenePath;
5     private string _componentsDirPath;
6
7     public SceneFactory(
8         string scenePath,
9         string componentsDirPath) { ... }
10
11     public Scene LoadResources() { ... }
12     private bool CompleteComponentType(string typeName) { ... }
13     private void CompleteComponentInstance(Component instance)
14     { ... }
15     private ComponentConfiguration? CreateConfig(string typeName)
16     { ... }
17
18     private string GetComponentTypePath(string typeName) =>
19         "{_componentsDirPath}/{typeName}";
20 }

```

4.1.5 SVG Loader

Listing 4.5 SkiaSvgLoader class

```

1 public class SkiaSvgLoader : ISvgLoader {
2     private readonly Dictionary<string, SvgDocument>
3         _loadedDocuments = [];
4
5     public void LoadDocuments(string componentPath) { ... }

```

```

6     public Dictionary<string, SvgDocument> GetDocuments() =>
7         _loadedDocuments;
8
9     public Dictionary<string, SKSvg> CompileSvgs() { ... }
10    public void ResizeDocuments(Vector2 desiredSize) { ... }
11 }

```

4.1.6 Configuration Loader

Listing 4.6 YamlConfigurationLoader class

```

1 public class YamlConfigurationLoader : IConfigurationLoader {
2     public record struct ConfigDto() {
3         public required string Name;
4         public string Description = "";
5         public string ImageSize = "";
6         public uint Pins = 0;
7         public List<string> PinNames = [];
8     }
9
10    public ComponentConfiguration? LoadConfig(
11        string configPath) { ... }
12
13    private ComponentConfiguration CreateConfig(
14        ConfigDto configDto,
15        string configPath) { ... }
16
17    private List<PinPrototype> CreatePins(
18        ConfigDto configDto) { ... }
19 }

```

4.1.7 Scene Loader

Listing 4.7 YamlSceneLoader class

```

1 public class YamlSceneLoader : ISceneLoader {
2     public class ComponentsMap : Dictionary<
3         string, SceneDto.ComponentDto>;
4     public class TypesMap : Dictionary<string, ComponentsMap>;
5     public class NodesList : List<List<string>>;
6
7     public record SceneDto {
8         public required TypesMap Components { get; init; } = [];
9         public required NodesList Nodes { get; init; } = [];
10
11         public record ComponentDto {
12             public string Position = "0 0";
13             public float Rotation = 0;
14             public string Scale = "1 1";
15
16             public Dictionary<string, object>? ExtraProps {
17                 get; set;
18             }
19         }
20     }
21 }

```

```

22     private SceneDto? _sceneDto;
23     private readonly Dictionary<string, Type> _typesByName = [];
24     private readonly Dictionary<string, Component>
25         _instancesByName = [];
26     private readonly Dictionary<string, List<Component>>
27         _instancesByType = [];
28
29     public Dictionary<string, Component> InstantiateComponents()
30     { ... }
31
32     private Type? MapComponentType(string typeName) { ... }
33
34     private Transform ParseTransform(
35         SceneDto.ComponentDto compDto) { ... }
36
37     private void CreateComponentInstance(
38         string typeName,
39         string componentName,
40         SceneDto.ComponentDto componentDto) { ... }
41
42     private void HandleExtraProps(
43         Component component,
44         Dictionary<string, object>? properties) { ... }
45
46     public List<ElectricalNode> InstantiateNodes() { ... }
47     public void LoadFile(string scenePath) { ... }
48     public List<string> GetComponentTypeNames() { ... }
49     public List<Component> GetComponentsOfType(
50         string typeName) { ... }
51 }

```

4.2 Main project with Avalonia UI

4.2.1 Program.cs, App.axaml and MainWindow.axaml

4.2.2 MainController static class

Listing 4.8 MainController class

```

1 public static class MainController {
2     public static CommandLineArguments Arguments {
3         get; private set; } = null!;
4     public static MainWindow MainWindow { get; private set; } = null!;
5     public static Scene Scene { get; private set; } = null!;
6     public static Stopwatch GlobalTimer { get; private set; } = null!;
7     public static ILogger Logger { get; private set; } = null!;
8
9     public static event Action? AppInitializedEvent;
10    private static bool _isInitialized = false;
11
12    public static void InitApp(
13        IClassicDesktopStyleApplicationLifetime desktop) { ... }
14
15    private static ILogger InitLogger(
16        string path,
17        LogLevel verbosity) { ... }

```

```

18
19     private static void InitScene() { ... }
20     private static void InitArduino() { ... }
21 }

```

4.2.3 ComponentControl templated control

Adjust Alt key behavior

Listing 4.9 ComponentControl class

```

1 public class ComponentControl : Control {
2     private Component _component = null!;
3     private bool _pressHeld = false;
4
5     public static readonly DependencyProperty<ComponentControl,
6         Component > ComponentProperty =
7         AvaloniaProperty.RegisterDirect<ComponentControl,
8             Component>(nameof(Component), o => o.Component,
9             (o, v) => o.Component = v);
10
11     public Component Component {
12         get => _component;
13         set => SetAndRaise(ComponentProperty,
14             ref _component, value);
15     }
16
17     public ComponentControl() { ... }
18
19     protected override void OnPropertyChanged(
20         AvaloniaPropertyChangedEventArgs change) { ... }
21
22     private void HandleComponentChanged(
23         AvaloniaPropertyChangedEventArgs change) { ... }
24
25     public override void Render(DrawingContext context) { ... }
26
27     protected override void OnPointerPressed(
28         PointerPressedEventArgs e) { ... }
29
30     protected override void OnPointerReleased(
31         PointerReleasedEventArgs e) { ... }
32
33     protected override void OnAttachedToVisualTree(
34         VisualTreeAttachmentEventArgs e) { ... }
35
36     protected override void OnDetachedFromVisualTree(
37         VisualTreeAttachmentEventArgs e) { ... }
38
39     private void OnGlobalKeyUp(object? _, KeyEventArgs e) { ... }
40
41     private void InvalidateVisual(SKSvg _) => InvalidateVisual();
42 }

```

4.2.4 SvgDrawOp drawing operation

Listing 4.10 SvgDrawOp class

```

1 public class SvgDrawOp(SKSvg Svg, SKMatrix Transform) :
2     ICustomDrawOperation {
3
4     public Rect Bounds =>
5         Transform.MapRect(_hitBox).ToAvaloniaRect();
6
7     private SKRect _hitBox => Svg.Picture!.CullRect;
8
9     public void Dispose() { ... }
10    public bool Equals(ICustomDrawOperation? other) => false;
11    public bool HitTest(Point p) { ... }
12    public void Render(ImmediateDrawingContext context) { ... }
13 }

```

4.2.5 CircuitCanvas user control

4.3 IPC Adapter project

4.3.1 Event struct

Listing 4.11 Event struct

```

1 public enum EventType { Unknown, Write, PinMode, Reboot }
2
3 public struct Event {
4     public const int MAX_ARGS = 3;
5
6     public EventType Type;
7     public long TimestampMicros;
8     public long LocalVirtualTime;
9     public int[] Args = new int[MAX_ARGS];
10
11     public Event() { }
12 }

```

4.3.2 IIpcAdapter interface

Listing 4.12 IIpcAdapter interface

```

1 public interface IIpcAdapter {
2     private const int RECONNECT_DELAY = 3000;
3     public bool IsConnected { get; }
4     public event Action<Event> ReceivedEventEvent;
5
6     public void SendEvent(Event @event);
7
8     public Task<bool> TryConnectAsync();
9     public void Disconnect();
10
11     public async Task ConnectAsync() {
12         while (!await TryConnectAsync()) {
13             await Task.Delay(RECONNECT_DELAY);
14         }
15     }
16 }

```

```

15     }
16
17     public void Reconnect() {
18         Disconnect();
19         Task.Run(ConnectAsync);
20     }
21 }

```

4.3.3 TextEncoder class

Listing 4.13 TextEncoder class

```

1 public class TextEncoder : IEncoder
2 {
3     private const int COMMON_PARTS = 3;
4     private const char PART_DELIMETER = ':';
5
6     private const string COMMON_PART_FORMAT = "{0:D12}:{1:D7}:";
7     private const string WRITE_FORMAT = "W:{0:D2}:{1:D1}";
8     private const string PINMODE_FORMAT = "P:{0:D2}:{1}";
9
10    public static string EncodeWrite(Event e) =>
11        string.Format(WRITE_FORMAT, e.Args[0], e.Args[1]);
12
13    public static string EncodePinMode(Event e) =>
14        string.Format(PINMODE_FORMAT, e.Args[0], (char)e.Args[1]);
15
16    public string EncodeEvent(Event @event) { ... }
17
18    public bool TryDecodeEvent(
19        string message,
20        out Event @event) { ... }
21 }

```

4.4 TCP Adapter project

4.4.1 TcpClient class

Listing 4.14 TcpClient class

```

1 public class TcpClient {
2     private const int RECONNECT_DELAY = 3000;
3
4     public bool Connected => _client?.Connected ?? false;
5     public readonly IPEndPoint Endpoint;
6
7     private const string LOCAL_HOST = "127.0.0.1";
8     private const uint BUFFER_SIZE = 4096;
9
10    private System.Net.Sockets.TcpClient? _client = new();
11    private NetworkStream? _stream = null;
12    private byte[] _buffer = new byte[BUFFER_SIZE];
13    private Channel<string> _sendQueue =
14        Channel.CreateUnbounded<string>();
15 }

```



```

16     private CancellationTokenSource _cts;
17     private ILogger? _logger;
18
19     public event Action<string> ReadBytesEvent;
20     public event Action ReachedEof;
21
22     public TcpClient(int port) { ... }
23     public TcpClient(string ip, int port) { ... }
24
25     public async Task<bool> TryConnectAsync() { ... }
26     public void Disconnect() { ... }
27     public async Task SendMessageAsync(string message) { ... }
28     public TcpClient SetLogger(ILogger? logger) =>
29         _logger = logger;
30     private async Task LoopReadAsync() { ... }
31     private async Task LoopWriteAsync() { ... }
32 }

```

4.4.2 TcpMessageHandler class

Listing 4.15 TcpMessageHandler class

```

1 public class TcpMessageHandler : IIpcAdapter {
2
3     public char Delimiter => ',';
4     public bool IsConnected => _client.Connected;
5
6     public event Action<Event>? ReceivedEventEvent;
7
8     private TcpClient _client;
9     private IEncoder _encoder;
10    private ILogger? _logger;
11    private StringBuilder _messageBuilder = new();
12
13    public TcpMessageHandler(
14        TcpClient client,
15        IEncoder encoder,
16        ILogger? logger = null) { ... }
17
18    public async Task<bool> TryConnectAsync() =>
19        await _client.TryConnectAsync();
20
21    public void SendEvent(Event @event) { ... }
22    public void Disconnect() => _client.Disconnect();
23    private void OnBytesRead(string bytechunk) { ... }
24    private void HandleMessage(string message) { ... }
25 }

```

4.5 Shared Memory Adapter project

4.5.1 CircularBuffer class

Listing 4.16 CircularBuffer class

```

1 public class CircularBuffer {

```

```

2 public const long PRODUCER_PTR_LOCATION = 0;
3 public const long CONSUMER_PTR_LOCATION = 8;
4 public const long BUFFER_LOCATION = 16;
5
6 private MemoryMappedViewAccessor _mappedRegion;
7
8 public long BufferSize => _mappedRegion.Capacity - BUFFER_LOCATION;
9 public long BytesAvailable => BufferSize - BytesFilled;
10 public long BytesFilled =>
11     Math.Abs(ProducerOffset - ConsumerOffset);
12
13 public bool IsEmpty => ConsumerOffset == ProducerOffset;
14 public long ProducerOffset { get; set; }
15 public long ConsumerOffset { get; set; }
16
17 public CircularBuffer(
18     MemoryMappedFile mappedMemory,
19     long offset,
20     long size) { ... }
21
22 public bool Write(ReadOnlySpan<byte> data) { ... }
23 public bool Write(string data) =>
24     Write(Encoding.ASCII.GetBytes(data));
25
26 public byte[]? ConsumeUntil(byte delimiter) { ... }
27
28 private bool FindDelimiter(
29     byte delimiter,
30     out long foundAtOffset) { ... }
31
32 private long ShiftOffset(long offset, long shift) =>
33     (offset + shift) % BufferSize;
34 }

```

4.5.2 TwoWayBuffer class

Listing 4.17 TwoWayBuffer class

```

1 public class TwoWayBuffer : IIpcAdapter {
2     public const string POSIX_SHARED_MEMORY_PATH = "/dev/shm/";
3     public const char MESSAGE_DELIMITER = ',';
4
5     public bool IsConnected { get; private set; }
6     public event Action<Event>? ReceivedEventEvent;
7
8     private string _memoryName;
9     private int _memorySize;
10
11     private MemoryMappedFile? _mappedMemory;
12     private CircularBuffer? _consumingBuffer;
13     private CircularBuffer? _producingBuffer;
14     private IEncoder _encoder;
15     private ILogger? _logger;
16
17     public TwoWayBuffer(
18         string name,
19         int pages,

```

```

20         IEncoder encoder,
21         ILogger? logger = null) { ... }
22
23     public void SendEvent(Event @event) { ... }
24     public Task<bool> TryConnectAsync() => Task.Run(TryConnect);
25     private bool TryConnect() { ... }
26     public void Disconnect() { ... }
27     private MemoryMappedFile OpenSharedMemory(string name)
28         { ... }
29     private void ListenForRemoteEvents() { ... }
30 }

```

5 User documentation

5.1 Compiling and running the simulator with an Arduino sketch

5.1.1 Repository file structure

For this section it is useful to know, where are the important files located within the source code repository. For the full description of the repository directories, see A.1. These are some of the more noteworthy directories and files located in the root of the repository:

backend/ contains source code of the backend simulation. The main `CMakeLists.txt` file is located here. It is also recommended to put the build directory here. The `CMakeLists` specifying the build targets and the `main()` function can be found within `backend/executable`.

frontend/ contains source code of the frontend GUI and component management. The `Gui.sln` solution file is located here. The `Program.cs` with the `Main()` function is located within the `frontend/Main` directory. The build directory will be also located here, if not stated otherwise. The components definition directory is on the path `frontend/ComponentManagement/Components`. This path must be specified using `-c <path>` command line argument, if `frontend` is not the current working directory.

tests/ contains example sketches and scenes used for testing both the backend and frontend.

preprocess.sh, preprocess.bat shell scripts performing basic sketch preprocessing using `arduino-cli`.

5.1.2 Build process and usage examples

First make sure you comply with the software prerequisites from the section 0.2. Make sure your Boost libraries can be found by the CMake metabuild system from your development environment. For further information, please follow the Boost's getting started guide [9]. After that, we can preprocess the sketch, compile it with the simulator, and run the simulator together with the frontend.

Backend process

1. Run `preprocess.sh` with the sketch directory. On Windows, use the `preprocess.bat` script instead. For example:

```
1 | ./preprocess.sh tests/buttons.
```

2. Build `backend/CMakeLists.txt`. Available targets are:

arduino-shared-memory for IPC over shared memory

arguino-tcp for IPC over TCP connection

arguino is an alias for the shared memory version (Default and preferred version).

3. Run ¹

```
1 | <build_dir>/executable/<build_target>
```

Frontend process

1. Build frontend/gui.sln.
2. Run the Gui executable² with the desired scene. For example, executing from the frontend directory:

```
1 | <build_dir>/Gui -s ../tests/buttons.scene.yaml
```

If you built the TCP version of the backend, be sure to add the flag **-i Tcp**.

For frontend and backend to communicate properly, it is necessary that they connect to the same IPC object. For TCP it is the port (**-p** option for both programs), and for shared memory it is the shared memory's object name (**--shm-name** option for both programs). These default to the same value for both of the aforementioned options, port 8888 and memory named **Arguino-ipc**.

5.1.3 Command-Line Arguments

Listed below are all the available command line options for the backend and frontend executables. Default values, if defined, are specified in the square brackets. Expected arguments are in the angle brackets. For arguments accepting only specific enumeration of values, all available values are listed in the angle brackets separated by a vertical bar (|). In square brackets is the default value used, when the option is omitted.

Backend

--log-simulator <path> Path to the log file for simulator events. [./core.log]

--log-ipc <path> Path to the log file for Inter-Process Communication (IPC). [./core_ipc.log]

-v, --verbosity <Debug | Info | Warning | Error> Verbosity level of the log files. [Info]

¹On Windows, the **.exe** extension is usually appended, i.e. **.../<build_target>.exe**.

²**Gui.exe** on Windows

Frontend

-s, --scene <path> Path to the .yaml scene definition file. [./scene.yaml]
-c, --components <path> Path to the directory containing component definitions. [./ComponentManager/Components]
-i, --ipc <None | Tcp | Shmem> Inter-process communication technology utilized for connecting with the simulator. [Shmem]
--log-circuit <path> Path to the log file for general circuitry events. [./frontend.log]
--log-ipc <path> Path to the log file for IPC events. [./frontend_ipc.log]
-v, --verbosity <Debug | Info | Warning | Error> Verbosity level of the log files. [Info]

Common for TCP usage

-p, --port <int> TCP port to listen to. [8888]

Common for shared memory usage

--shmem-name <string> Name of the memory-mapped file. [Arguino-ipc]
--shmem-size <int> Size of the memory-mapped region per circular buffer, specified in pages (usually $page = 4096B$). [1]

5.2 Defining components and scenes

Extensibility of the frontend through custom components and circuits is an important part the project. Here are necessary steps and requirements to create and use new components.

5.2.1 Required file structure

All of the components must reside in one directory, referred to as *definitions directory*. Each component is represented by exactly one directory, referred to as *component directory*, placed in the definitions directory. The definitions directory must be supplied through the **-c** command line argument. These files must be present in the component directory:

- C# behavior script
- YAML component configuration
- One or more .svg images

The yaml file must have the same name as the component directory, i.e. `MyComponent/MyComponent.yaml`. The SVGs will be referenced in behavior script by their file name, excluding the .svg extension. For example `image.svg` will be referred to as `image`.

Notes on used YAML notation

In the next subsections, we will describe the expected structures of the YAML files using this notation:

- angle brackets must be replaced by proper parameters, i.e. `name: <string>`
- properties with question mark are optional, i.e. `optional?: ...`
- Vector2 is represented as a string of 2 float values separated by a space, i.e. `vector: 7.27 6.9`

5.2.2 Component configuration file specification

This file defines common properties of the custom component type.

Listing 5.1 YAML configuration file structure

```
1 name: <string>
2 description?: <string>
3 imageSize?: <Vector2>
4 pins?: <uint>
5 pinNames?:
6   - <string>
7   - ...
8   - ...
```

name Name of the component type as displayed in the UI. (not used yet)

description Brief description of the component as displayed in the UI. (not used yet)

imageSize Desired size of the component's sprites in pixels for default zoom level of the canvas. Defaults to <50, 50>.

pins Number of unnamed pins. These pins are identified by their 0-based index in both the scene definition and in the component logic.

pinNames List of unique names for each pin. These pins are identified by their given name in both the scene definition and in the component logic.

Listing 5.2 Component configuration example: `Led.yaml`

```
1 name: Led
2 description: Light emitting diode
3 imageSize: 50 50
4 pinNames:
5   - anode
6   - cathode
```

5.2.3 Component behavior script specification

How does the component interact with the rest of the circuit is described by the C# class with the named the same as the component type. This class must inherit from the abstract `Component` class. The file containing the class must be added to the `ComponentManagement.csproj` project. In order to compile, the user's class must also define a constructor like this:

```
1 public MyComponent(string definitionPath)
2 : base(definitionPath) { ... }
```

However, no code is required in the body of the constructor. The constructor can be used for initialization of properties known before the scene is created. The `Component` class provides most of the necessary API.

Component functions and properties

virtual void OnInitialized() Gets called when all of the component instances have been initialized and scene created.

virtual void OnPinConnected(Pin) Gets called when a component's pin connects to an electric node. The connected pin is passed as argument.

virtual void OnPinDisconnected(Pin) Gets called when a component's pin disconnects from an electric node. The disconnected pin is passed as argument.

virtual void OnPinStateChanged(Pin) Gets called when a component's pin detected a change in digital state of a node it has been connected to. It can be thought of as changing input value on the pin. Pins set to `WriteOnly` mode by design don't detect this change. The detected pin is passed as argument.

virtual void OnControlPress(Vector2) Gets called when the component has been pressed on the circuit canvas. The cursor position at the time of the press is passed as argument.

virtual void OnControlRelease Gets called when the component has been released on the circuit canvas.

void UpdateSprite(string) Changes the .svg image drawn to the canvas. The new image is given by the svg name specified in the argument.

Pin? GetPin(string) Returns a `Pin` object with the name given in the argument. The name must be defined in the component's `namedPins` list. Returns null if no such pin exists.

Pin? GetPin(uint) Returns a `Pin` object with the id given in the argument. The allowed range of Ids is from 0 to `pins + namedPins.Length`. With unnamed pins starting from the id 0 and named pins starting from the id `pins`. Returns null if the pin id is out of range.

Pin functions and properties

uint Pin.Id Id of the pin, as described in the `GetPin(uint)` section.

string Pin.Name Name of the pin as described in the component configuration.

bool Pin.IsDriving True if the pin is driving. That is, if it actively forces the digital state on the connected node to be high. If the node has at least 1 driving pin, it's digital state will be high. It will be low otherwise.

bool Pin.IsHigh True if the digital state of the connected node is high.

bool Pin.IsLow True if the digital state of the connected node is low.

bool Pin.IsWriteOnly Write only pins don't react to changes in the connected node's digital state.

bool Pin.IsReadOnly Read only pins cannot be set to driving.

bool Pin.IsConnected Whether is the pin connected to any electrical node.

void Pin.MakeReadOnly() Sets the pin to read only mode. Read only pins cannot be set to driving.

void Pin.MakeWriteOnly() Sets the pin to write only mode. Write only pins don't react to changes in the connected node's digital state.

void Pin.MakeGeneral() Sets the pin to general mode. General pins don't pose any restrictions to reading or writing.

void Pin.SetValue(bool) If the argument is true, it sets the pin to be driving. Otherwise, the pin set to not drive.

void Pin.SetHigh() Sets the pin to be driving. Same as `SetValue(true)`.

void Pin.SetLow() Sets the pin to be not driving. Same as `SetValue(false)`.

Listing 5.3 Button.cs

```
1 public class Button : Component {
2     private const string SPRITE_BUTTON_UP = "buttonUp";
3     private const string SPRITE_BUTTON_DOWN = "buttonDown";
4
5     private bool _isPushed = false;
6
7     public Button(string definitionPath) : base(definitionPath) {}
8
9     internal override void OnInitialized() {
10         UpdateSprite(SPRITE_BUTTON_UP);
11     }
12
13     public override void OnPinStateChanged(Pin pin)
14         => UpdatePins();
15
16     public override void OnControlPress(Vector2 cursorPosition) {
17         _isPushed = true;
```

```

18         UpdateSprite(SPRITE_BUTTON_DOWN);
19         UpdatePins();
20     }
21
22     public override void OnControlRelease() {
23         _isPushed = false;
24         UpdateSprite(SPRITE_BUTTON_UP);
25         UpdatePins();
26     }
27
28     private void UpdatePins() {
29         UpdatePin(GetPin(0)!);
30         UpdatePin(GetPin(1)!);
31     }
32
33     private void UpdatePin(Pin pin) {
34         Pin otherPin = GetPin(1 - pin.Id)!;
35
36         if (otherPin.IsHigh && _isPushed) {
37             pin.SetHigh();
38         }
39         else {
40             pin.SetLow();
41         }
42     }
43 }

```

In the example above, we have defined a behavior of a `Button` component. The component's configuration specifies two unnamed pins, and component's directory contains 2 SVG files, `buttonUp.svg` and `buttonDown.svg`. The pins are in the general mode, since we didn't specify their mode explicitly.

In the overridden `OnInitialized` we set the `buttonUp` image as the default sprite. In the overridden `OnControlPress` and `OnControlRelease` methods we update the internal state of the button: whether it is pressed or not. We change the actively displayed sprite accordingly.

`UpdatePins` is a helper method specific to this component. For both of the unnamed pins (retrieved by the `GetPin` method) it checks whether the value of the pin should be changed. A pin is set to be driving (`SetHigh` method), if the button is pushed, and the other pin reads a high value. We have to be careful, just because a pin reads a high value, it doesn't mean we want to set it to driving. Once a state of a node drops back to low, we want states of both pins drop back to low as well, regardless of whether the button is pushed or not. ³This update method is called everytime a button is pressed, released, or a value on of its pins changes (`OnPinStateChanged` override).

5.2.4 Defining scenes

The scene YAML file describes component instances present in the scene, and how are their pins connected together in nodes. The path of this file is supplied through the `-s` command line argument. There are no restriction to where the scene file has to be placed.

³Unless, of course, the node at the other side is also high. That means the pin was, and will stay driving

Listing 5.4 YAML scene file structure

```
1 components:
2   <Type name>:
3     <Instance name>:
4       position: <Vector2>
5       rotation?: <float>
6       scale?: <Vector2>
7     <Instance name>:
8       ...
9 nodes:
10  - [ <Instance name>.<Pin name>, ...]
11  - ...
12  - ...
```

components Dictionary of component type objects. These objects are dictionaries of component instances of that type. These instances define properties of the individual components.

Type name String. Name of the component type. This must match the component directory name.

Instance name String. Name of the component instance. It identifies the instance within the node definition and in the UI. It must be unique across all instances within the components dictionary.

position Coordinates on the circuit canvas, with origin in the upper left corner.

rotation Clockwise rotation angle in degrees on the circuit canvas.

scale Relative scaling along x and y axis on the circuit canvas.

nodes List of lists. The inner list represents all pins connected to the same electrical node. Its element is an instance-pin pair separated by a dot. Instance name must be present in the components dictionary. Pin name must be present in the component's configuration. In the case of unnamed pins, pin name is simply its index.

Listing 5.5 Scene example: minimal.scene.yaml

```
1 Components:
2   Led:
3     Led1:
4       Position: 100 100
5     Led2:
6       Position: 200 100
7   Button:
8     Btn:
9       Position: 100 200
10  Vcc:
11    Source:
12      Position: 125 300
13 Nodes:
14  - [Source.0, Btn.0]
15  - [Led1.cathode, Btn.1]
16  - [Led2.cathode, Btn.1]
```

The example scene in the listing 5.5 contains 2 LEDs, 1 button and 1 source of voltage. Source, which has one pin that is always driving, is connected to the 0th pin of the button, which reads a high value of the node. Both LEDs are connected to the 1st pin of the same button. Pressing the button will copy the digital state from pin 0 to pin 1, pin 1 becomes driving, and thus lights up the LEDs.

5.3 IPC interface

This section describes how to set up IPC communication of a custom frontend application with the backend simulator. The two processes can communicate together either via TCP connection or through shared memory. The only requirement for a TCP connection, is that the client on the frontend listens to the same port as the server on the backend. IPC over shared memory requires a few more considerations, so that the memory accesses match the memory layout of shared internal structure created by the backend. This structure is described in the subsection 5.3.3;

The processes communicate using concrete events with up to 3 integer arguments. These events are encoded into a human readable text messages, separated by a common delimiter. The messages are then sent over IPC.

5.3.1 Text message format

Individual messages are delimited by the semicolon character (;). The individual values within one message are delimited by the colon character (:). The integer values are written in decimal system, using the ASCII characters of the actual digits. The semantics of these values are better described in the chapter about architecture, subsection 2.2.3. The format of the message then looks as the following:

Listing 5.6 Format of an IPC text message

```
1 | <sim_time>:<lvt>:<event_type>:<arg1>:<arg2>:<arg3>;
```

sim_time Simulation time, 64-bit signed integer. Real time elapsed from the start of the backend's simulation represented in microseconds. The backend has no use for this value.

lvt Local virtual time, 64-bit signed integer. It describes an order in which the event occurred, relative to other events.

event_type A single character representing the type of the event. This type decides the action taken upon consuming the event. Recognized event types and their corresponding characters are described in the next subsection.

arg1, arg2, arg3 32-bit signed integers. Values have meaning depending on the event. Arguments not defined by the event type can be omitted. Make sure that the number of colons match the number of passed arguments.

For example, this next message encodes the **Write** event passing 2 arguments at the simulation time of 264 seconds:

Listing 5.7 Example of an IPC text message

```
1 | 000264997488:0000016:W:08:1;
```

Decide on whether to describe or remove the zero paddings.

5.3.2 Recognized event types

Write event

Signalizes digital value being written into a pin. If consumed by the backend, the value gets written to the internal **ArduinoState**. If consumed by the frontend, it should reflect the changes caused by the change of the pin's state. This event is produced by the backend on **digitalWrite** function call within the sketch.

Encoding character: W

Arguments:

- 1: Pin index** of the digital pin from the range 0–13.
- 2: Value** 0 for the low value, 1 for the high value.

Pinmode event

Signalizes change of a pin mode of a pin. If consumed by the backend the value gets written to the internal **Arduinostate**. If consumed by the frontend, it should change the behavior of the pin, such that only input pins are able to send the **Write** event. This event is produced by the backend on **pinMode** function call within the sketch.

Encoding character: P

Arguments:

- 1: Pin index** of the digital pin from the range 0–13.
- 2: Mode** 0 for the output mode, 1 for the input mode.

Reboot event

Produced by the backend, when the simulation has started over. Useful, when the backend process has be terminated and then executed again. At this point the frontend has to invalidate its current state and start again. Backend itself does no action upon consuming this event (it may force the reboot of the simulation in the future).

Encoding character: R

Arguments: None



Figure 5.1 Memory layout of the circular buffer

5.3.3 Shared memory specification

Locating memory mapped file

On Linux, the mapped memory region is orchestrated by a proper shared memory object. It can be accessed either by a syscall, or by reading the `\dev\shm\<shmem_name>` virtual file.

On Windows, the shared memory is emulated by the `Boost.interprocess` library via a memory mapped file. It can be located at the path specified by the `Common AppData` registry. For Windows Vista and newer, this path defaults to `C:\ProgramData`. On this path, a `boost_interprocess` directory is created. The memory mapped file, named the same as the shared memory object, can be found in one of its subdirectories. [10]

In either case, the name of the read memory mapped file or shared memory object has to match the name given to the mapped region via the `--shmem-name` flag of the backend command.

Memory layout

The mapped memory region is split into 2 equal sections, both containing the same circular buffer data structure. So for a mapped region of size n bytes, and for the start address of the region s_a , the second buffer is located on the memory address $s_a + n/2$ (n is always even, as it is a multiple of a page size). Let's denote the start address of a buffer s_b .

The buffer itself has 2 control variables stored within it. Both are 64-bit unsigned integers representing a byte offset from the beginning of the buffer's data. The first, located at s_b , is the offset at which the producer will write the next chunk of data into the buffer. The second, located at $s_b + 8$, is the offset at which the next chunk of data will be read from the buffer, if present. From the address $s_b + 16$ until the next buffer, or the end of the mapped region, the actual data are stored. That means the actual capacity of data stored within one buffer is $n/2 - 16$.

The data have no uniform size, structure or address alignment. The length of a message in the buffer must be deduced from the data within the message itself. In case of the text encoding, it is the message delimiter. A message can be located at the boundary of the data region. In that case the message is split so that every byte until the very end of the region is filled, and the rest is written at $s_b + 16$ again. There are no gaps between the messages.

The frontend process is responsible for consuming messages from the first buffer and producing them to the second one. As such, the consumer offset of the first buffer and the producer offset of the second buffer must be updated accordingly. The other two offsets must remain read only. The backend process has these roles reversed.

Conclusion

In the end we managed to develop two nontrivial applications. One for running and simulating the user's Arduino code. The other one for creating a interactive virtual environment, where the Arduino can be connected into a circuit with visual feedback. In chapter 2 of this thesis we described the thought process, project structure, and technologies used in order to achieve the goals we set for ourselves in section 0.2. Then we looked into more detail on how we implemented these technologies while avoiding many pitfalls in chapters 3 and 4.

While we implemented only a couple of most used functions from the Arduino library, and the user interface leaves a lot to be desired, we believe the current version of the product represents a solid foundation on which new functions, events, components and quality of life features can be laid down. One of the biggest hurdles was making sure that the two applications stay synchronized. We succeeded in that regard, by learning more of useful information about interprocess communication and approaches commonly used in distributed computing.

A big success also lies in the extensive component system developed in the frontend. The user can extend their component library just by following the specification in section 5.2. They can use custom vector graphics, and give life to their components thanks to the abstract **Component** class. Then they can create it into a scene together with Arduino running their code. All this while all the heavy work is done by the reflection and parsing machinery on the background. We would be excited to work on this component system, so that it brings even more possibilities to the user.

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A Attachments

A.1 Source code repository

The file attachment `thesis.zip` contains the repository with the source code in its entirety. The repository is equivalent with the public git repository found at the URL <https://github.com/TenIdiotZInternetu/Arguino> at the tag `add tag`.

The file structure of the repository is as follows:

`.vscode/` contains the files:

`launch.json` specifies backend debug configurations using the `cppdbg` VS Code extension.

`settings.json` specifies backend source and build directories using the `CMakeTools` VS Code extension.

`backend/` contains the source code for the backend application. Each directory corresponds to a single module described in the chapter 3, with the exception of two:

`libs/` contains the two utilities modules: `logger` and `timer`.

`.vscode/` contains the setup equivalent to the `.vscode` directory in the root of the repository.

Other files in this directory are:

`.clang-format` specification of the automatic code formatting.

`.gitignore` exclusion of build files.

`CMakeLists.txt` main metabuild specification for the backend. References `CMake` files in the module subdirectories.

`CMakePresets.json` build configurations using Clang on Linux and MSVC on Windows.

`vcpkg.json` specifies `Vcpkg` dependencies for the Boost libraries.

`docs/` Contains year project specification and other documentation:

`RP_Specifikacia.odt`, `RP_Specifikacia.pdf` final year project specification in Slovak language.

`specification-draft.md` rough version of the year project specification in English language. Contains the original vision and expectations for the project.

`component_addons_doc.md` contains specification of custom components and scenes configuration in the scope of the section 5.2.

`frontend/` contains the source code for the frontend application. Each directory corresponds to a single project described in the chapter 4. The remaining files are `.gitignore` and the `gui.sln` .NET solution file.

res/ contains assets used in source code unrelated context, e.g. screenshots.

tests/ contains example sketches and scenes used for testing both the frontend and the backend.

thesis/ contains source files for this document, e.g. **.tex** files, bibliography files and figures.

.gitignore excludes build and Visual Studio files

README.md describes the usage and build process of both the backend and the frontend in the scope of the section 5.1.

preprocess.sh shell script performing basic sketch preprocessing using **arduino-cli**. For use on Linux.

preprocess.bat same as **preprocess.sh** but for the use on Windows.